Pipeline - Benchmark summary

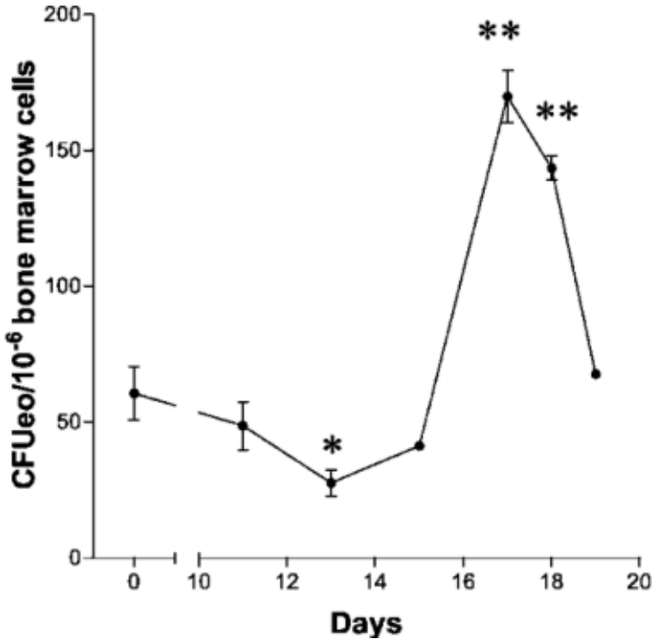
Model: gemini-3-pro-preview
Cases: 35
Successes: 29

nMAE (mean / median, n=28) 0.0519 / 0.0273
L1 normalized (mean / median, n=4) 0.0126 / 0.0146
Duration seconds (mean / median, n=29) 54.8 / 43.1
Series count mismatches (extracted vs GT): 0

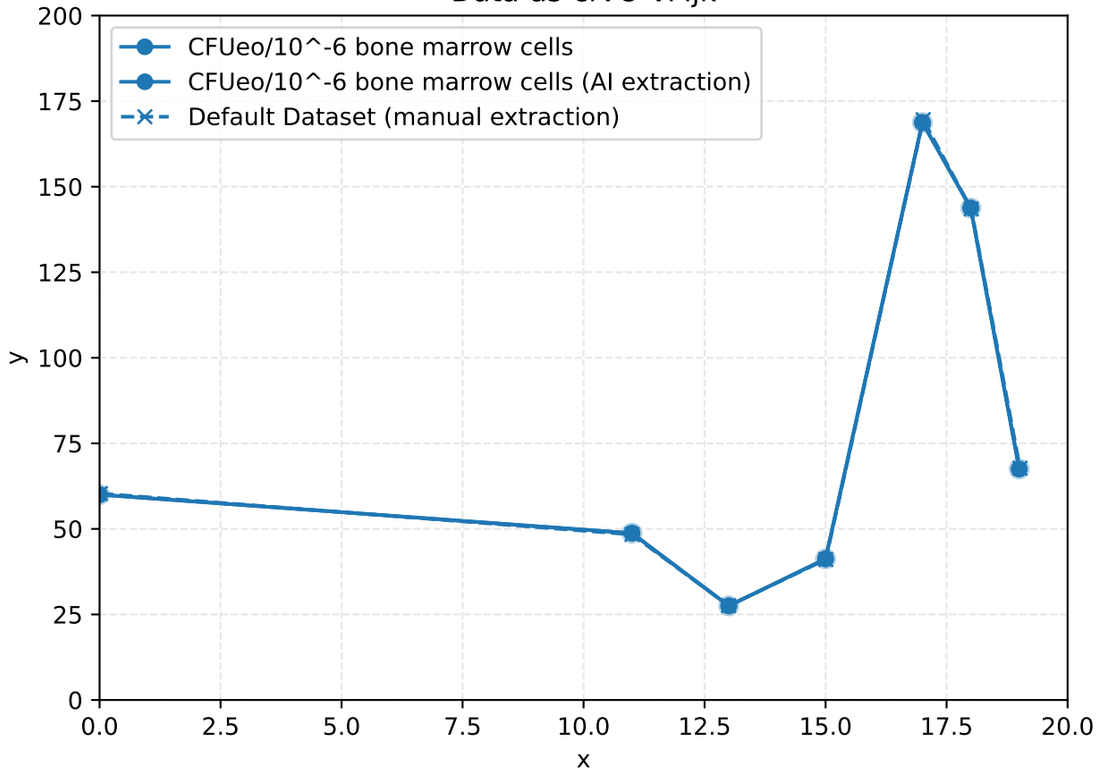
Pipeline - Benchmark summary - Prompt

```
 You are an expert system that extracts NUMERICAL data from 2D scientific plots. You must
return ALL visible series from the given image.  Rules: - Do not guess or invent data. -
Work in three internal steps (do NOT show them):   1) Read axes:      - Determine x_min,
x_max (detect log10 if any).      - Determine y_min, y_max (detect log10 if any).      -
List visible tick labels for x and y (as numbers).   2) Identify all visible data series:
- For discrete markers: read their positions and map to axis coordinates.      - For
continuous curves: sample >= 20 points (include local extrema, inflections, and uniform
samples).   3) Validate:      - Ensure x in [x_min, x_max], y in [y_min, y_max].      - For
linear axes, reject perfectly uniform sequences unless they exactly match tick labels.
- If uncertain about a point, omit it. Never invent.  Output format (STRICT JSON, no extra
text): {   "series": [     {       "label": "<legend name if visible else 'series_N'>",
"points": [         {"x": <float>, "y": <float>, "confidence": <float 0..1>},         ...
],       "axis": {         "x": {"min": <float>, "max": <float>, "scale": "linear"|"log10"},
"y": {"min": <float>, "max": <float>, "scale": "linear"|"log10"}       }     },     ...   ]
}  Additional rules: - Include ALL visible series. - Keep points sorted by increasing x. -
Use at least 20 points for continuous curves if readable; otherwise return all discrete
markers. - Never include explanations, placeholders, or text outside JSON.
```

Source image



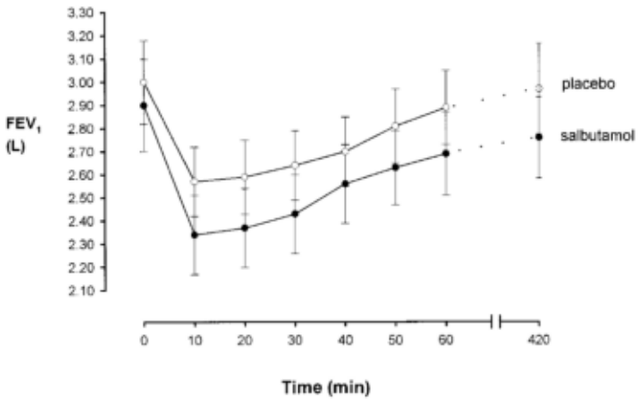
Data-as-cfVO-vMjk



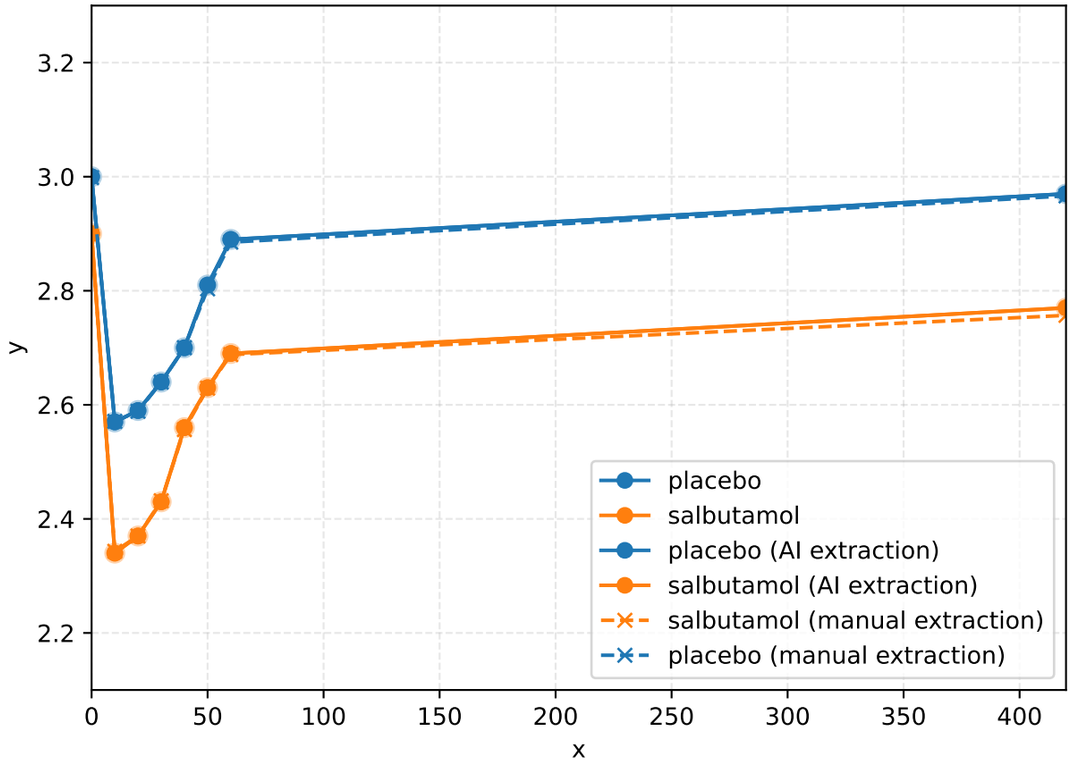
Metrics

Metric	Value
duration (s)	33
overall_nmae	0.002902
overall_integral_l1	2.126
overall_integral_l1_normalized	0.0007885
overall_x_misaligned	False
extracted_series_count	1
GT_series_count	1

Source image



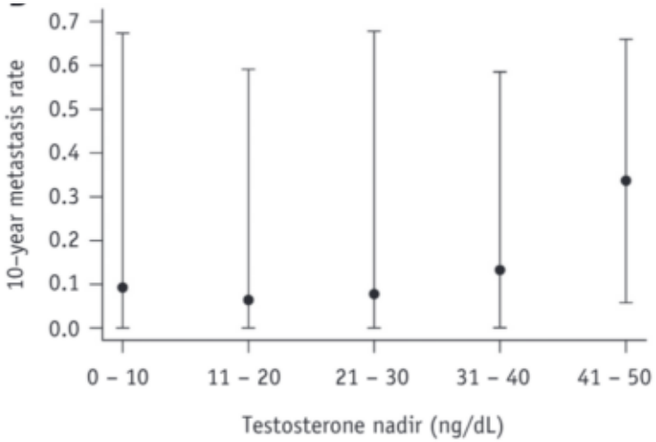
Data-as-8975-q7O8



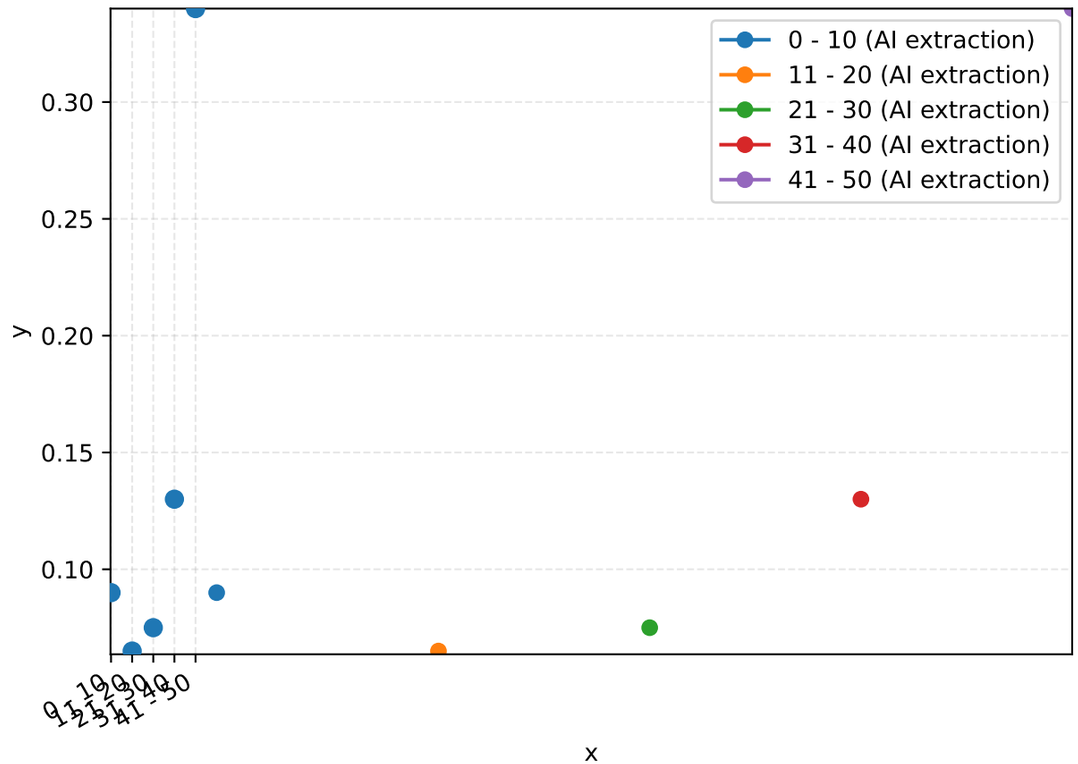
Metrics

Metric	Value
duration (s)	41
overall_nmae	0.006094
overall_integral_l1	2.263
overall_integral_l1_normalized	0.01081
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



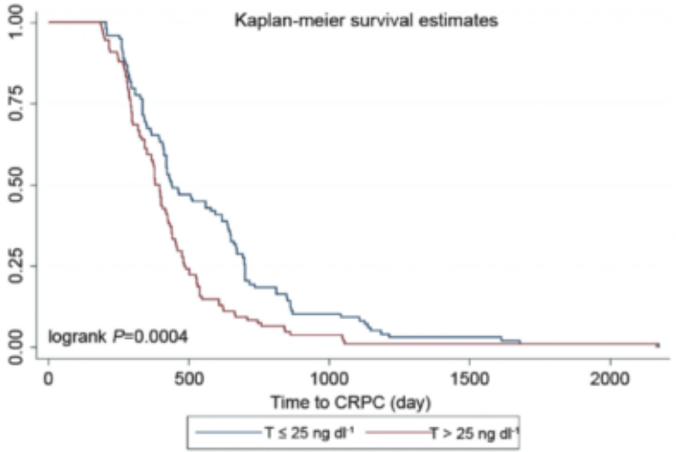
Data-ProstateCancer-10YearMetastasis-TestosteroneNadir-as-HUFM-k4RT



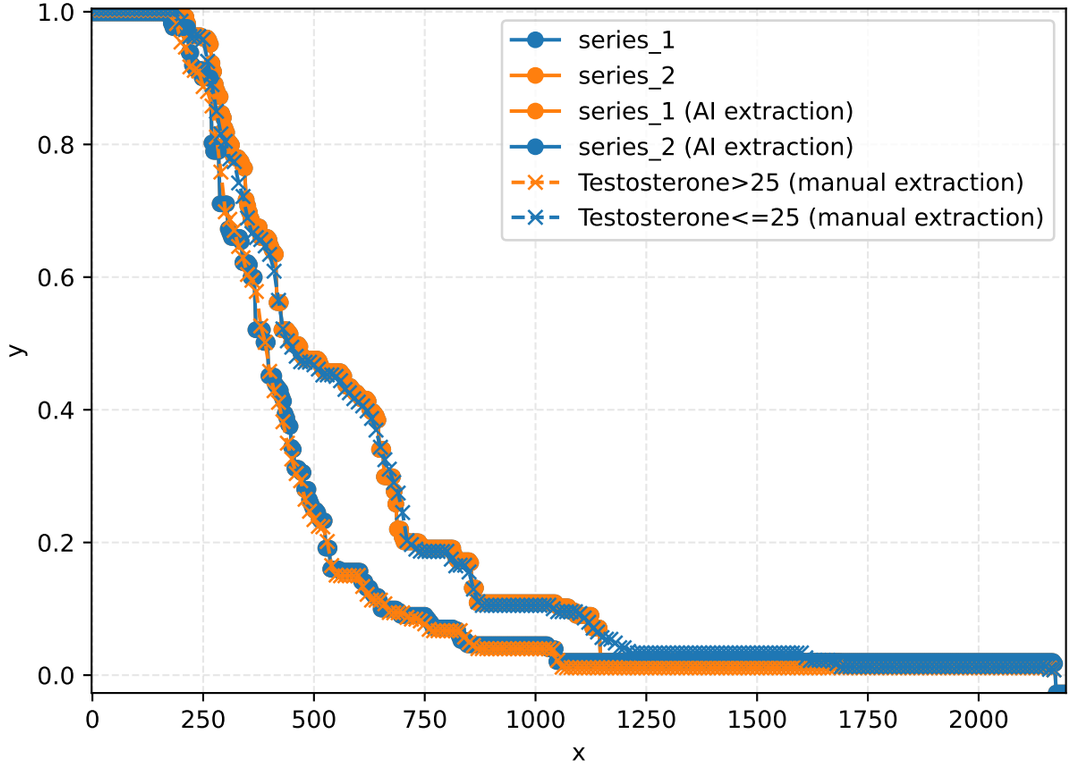
Metrics

Metric	Value
duration (s)	62
overall_nmae	0.00946
overall_integral_l1	nan
overall_integral_l1_normalized	—
overall_x_misaligned	True
extracted_series_count	5
GT_series_count	5

Source image



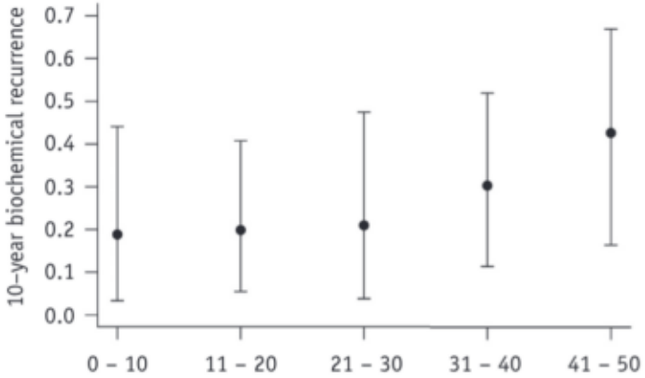
Data-ProstateCancer-TimeToCastrationResistance-CastrationLevel-as-G8iP-2p8H



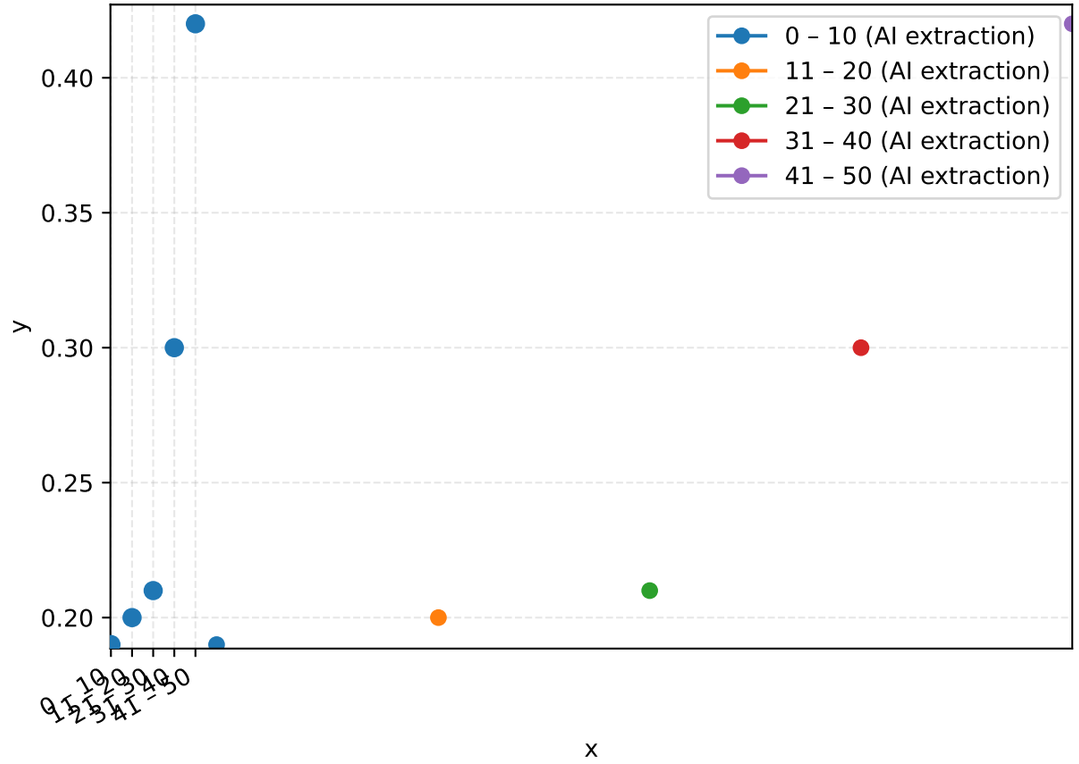
Metrics

Metric	Value
duration (s)	40
overall_nmae	0.01024
overall_integral_l1	19.44
overall_integral_l1_normalized	—
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



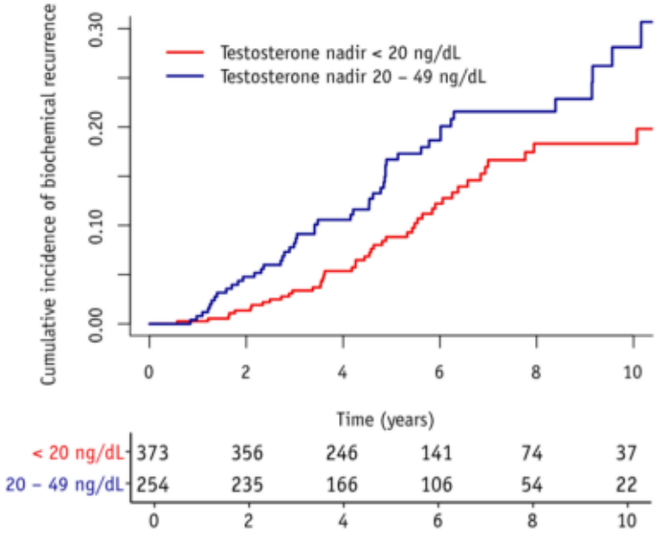
Data-ProstateCancer-10YearBiochemicalRecurrence-TestosteroneNadir-as-FRp3-CyQW



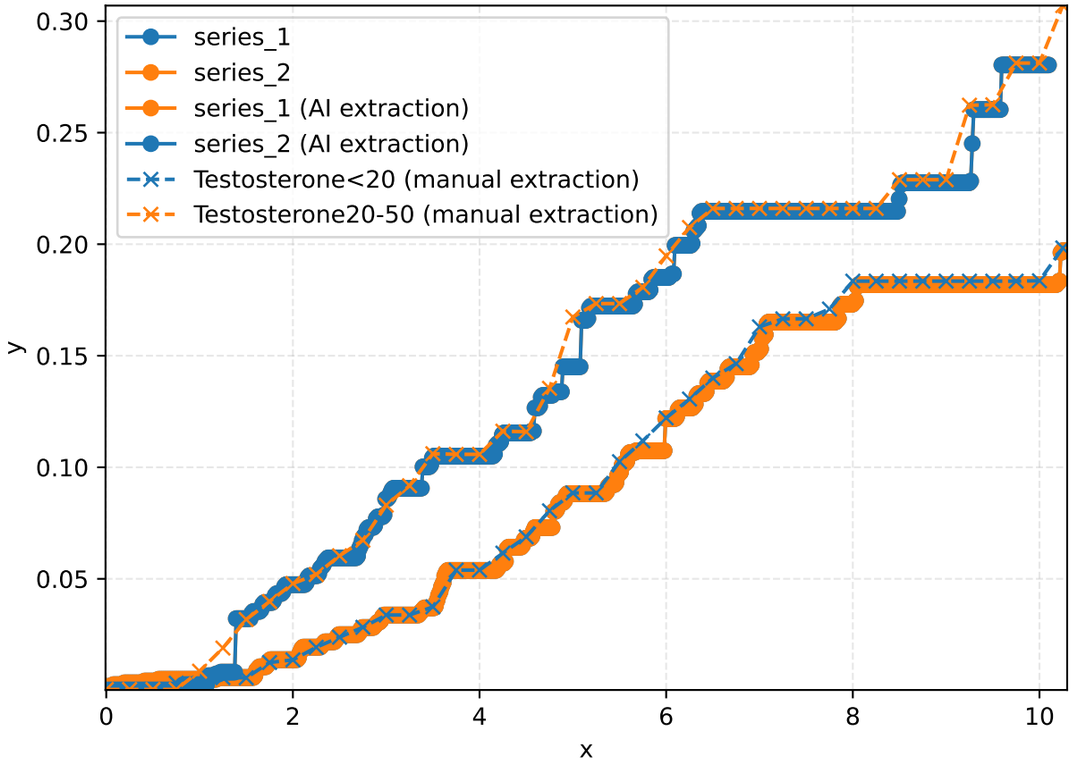
Metrics

Metric	Value
duration (s)	35
overall_nmae	0.01162
overall_integral_l1	nan
overall_integral_l1_normalized	—
overall_x_misaligned	True
extracted_series_count	5
GT_series_count	5

Source image



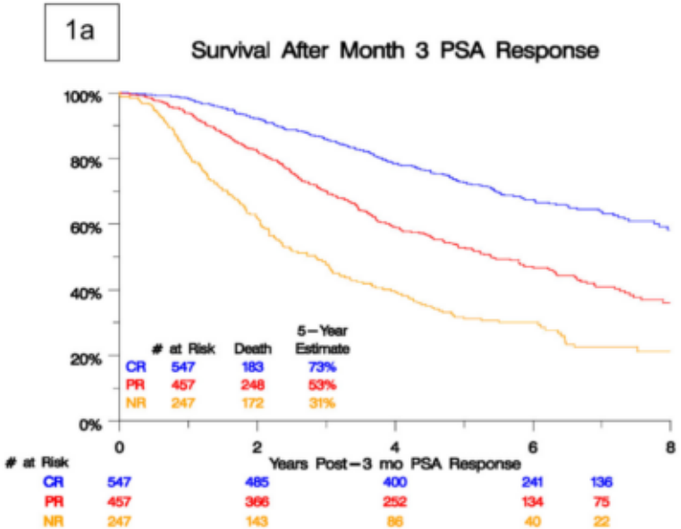
Data-ProstateCancer-BiochemicalRecurrence-TestosteroneNadir-as-kPUM-ja2E

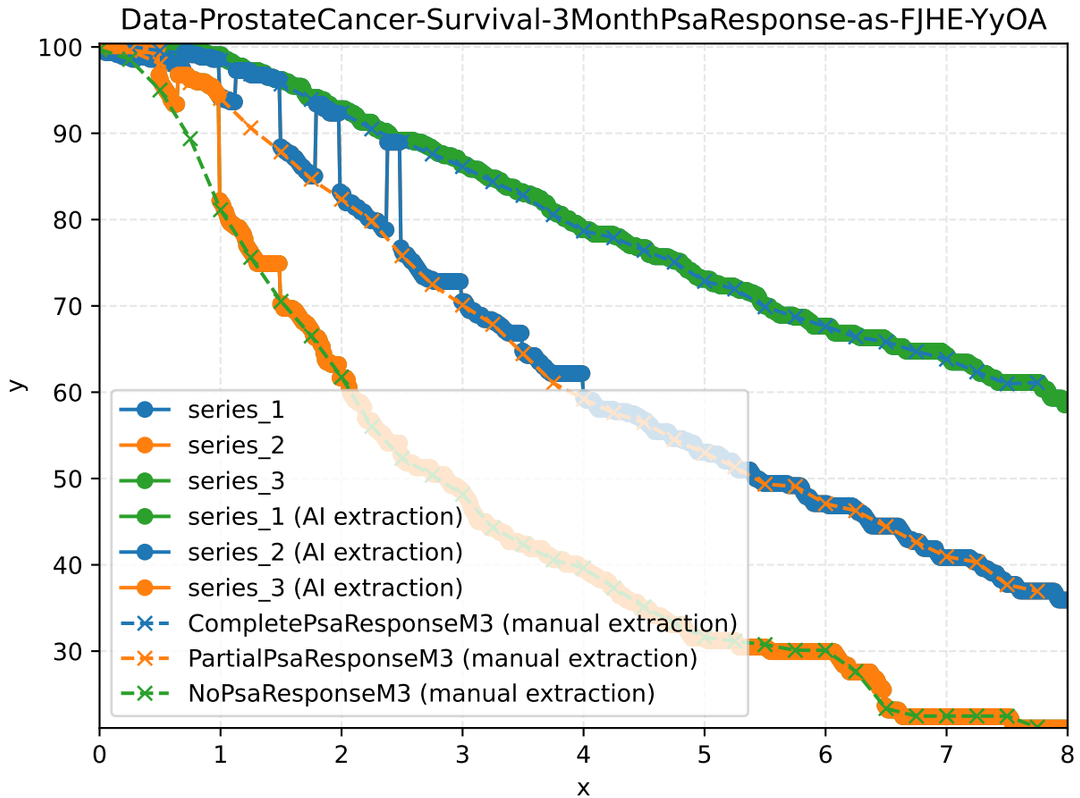


Metrics

Metric	Value
duration (s)	47
overall_nmae	0.01175
overall_integral_l1	0.02805
overall_integral_l1_normalized	—
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image

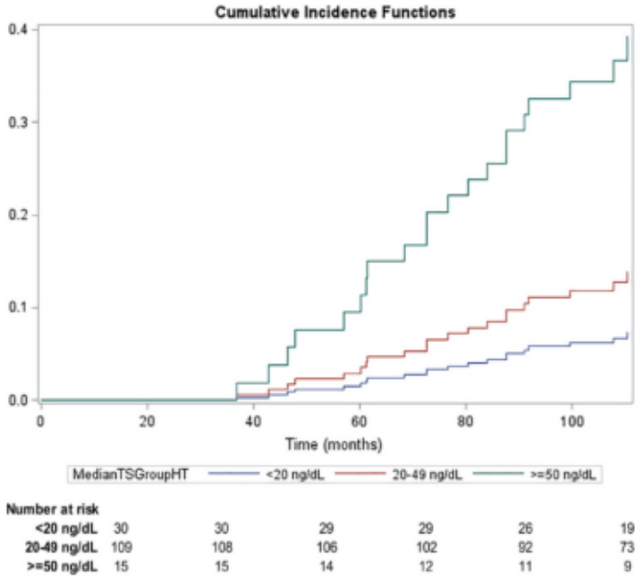




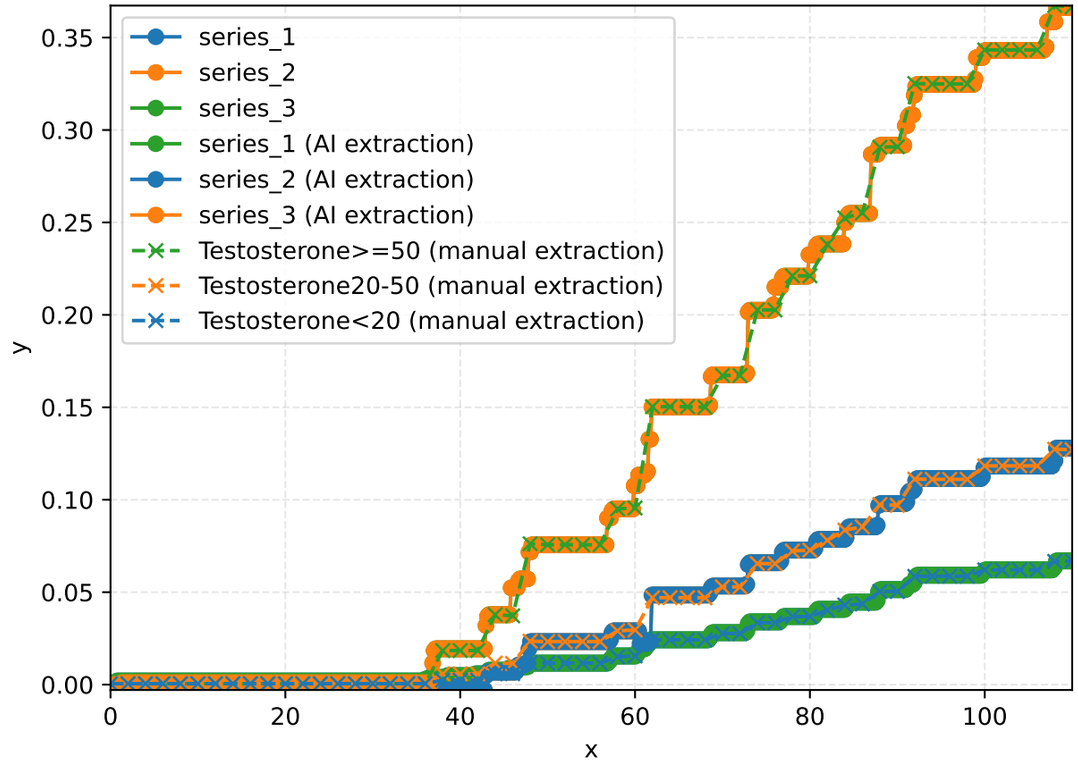
Metrics

Metric	Value
duration (s)	53
overall_nmae	0.01331
overall_integral_l1	6.403
overall_integral_l1_normalized	—
overall_x_misaligned	False
extracted_series_count	3
GT_series_count	3

Source image



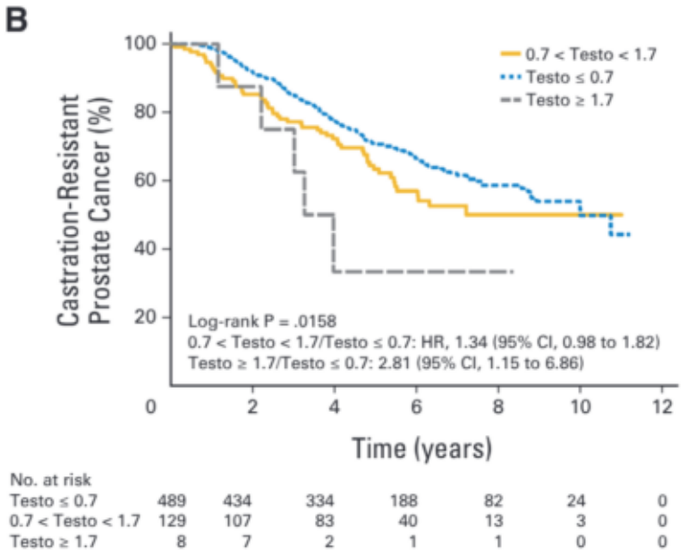
Data-ProstateCancer-TimeToBiochemicalFailure-TestosteroneMedian-as-V4ML-TC46



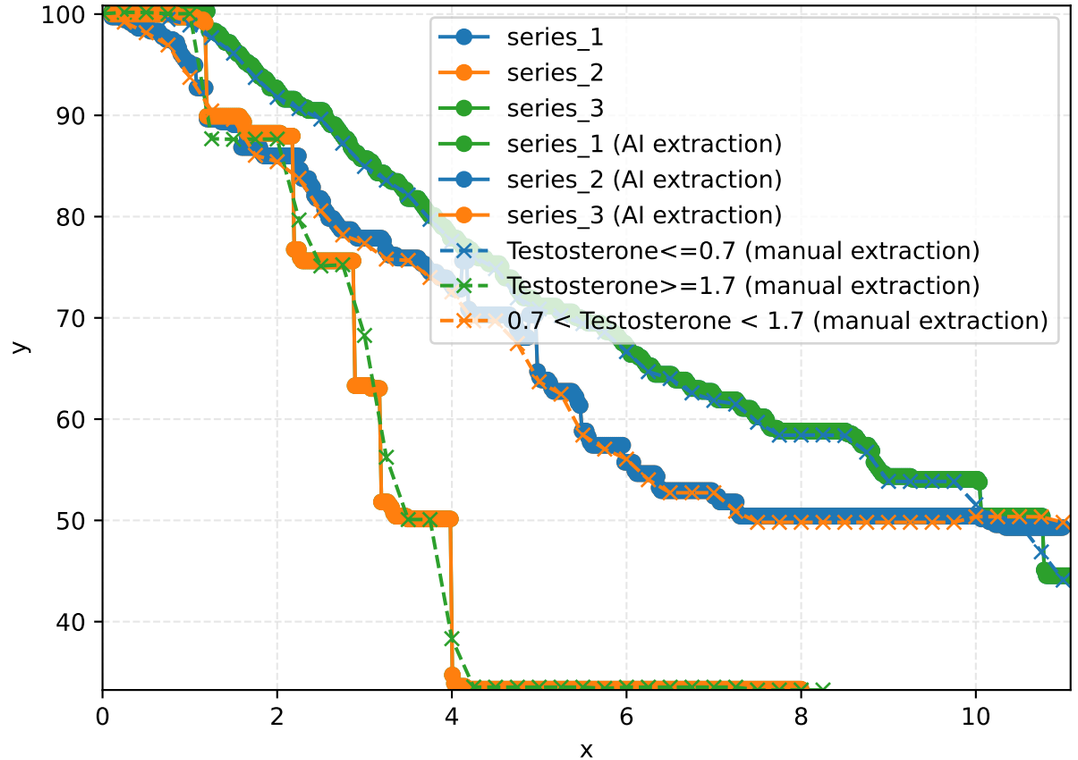
Metrics

Metric	Value
duration (s)	58
overall_nmae	0.01362
overall_integral_l1	0.2029
overall_integral_l1_normalized	—
overall_x_misaligned	False
extracted_series_count	3
GT_series_count	3

Source image



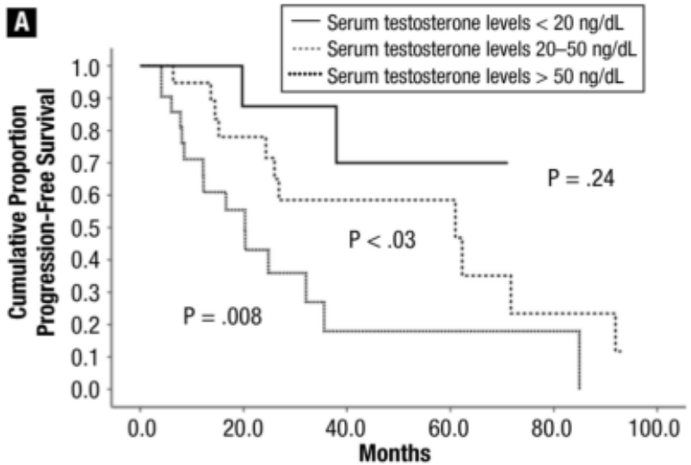
Data-ProstateCancer-TimeToCastrationResistance-TestosteroneNadir-as-KPBC-NCfw



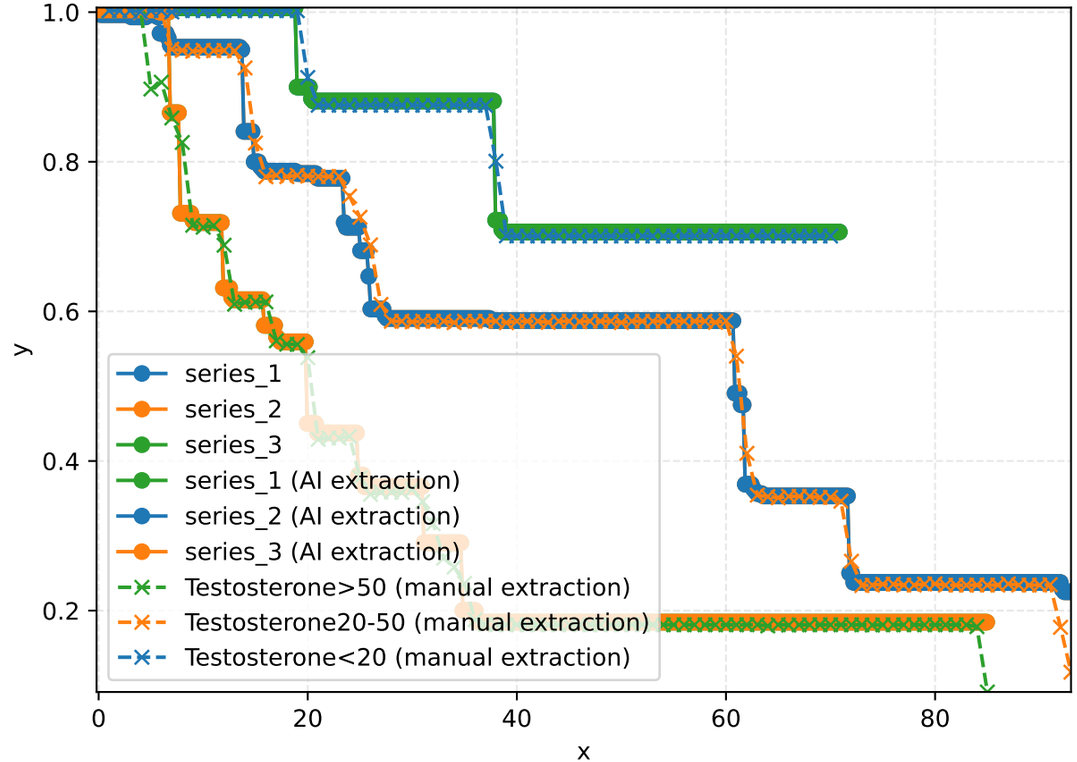
Metrics

Metric	Value
duration (s)	56
overall_nmae	0.01403
overall_integral_l1	7.657
overall_integral_l1_normalized	—
overall_x_misaligned	False
extracted_series_count	3
GT_series_count	3

Source image



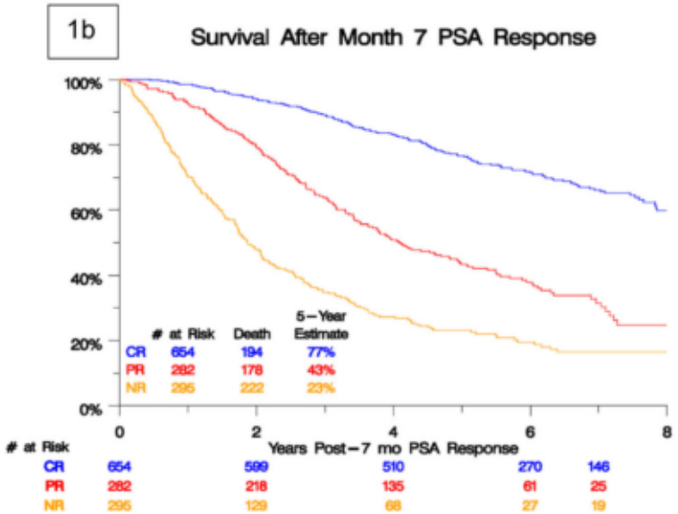
Data-ProstateCancer-CastrationLevel-TimeToProgression-LocallyAdvanced-Metastatic-as-aWvR-2wlo

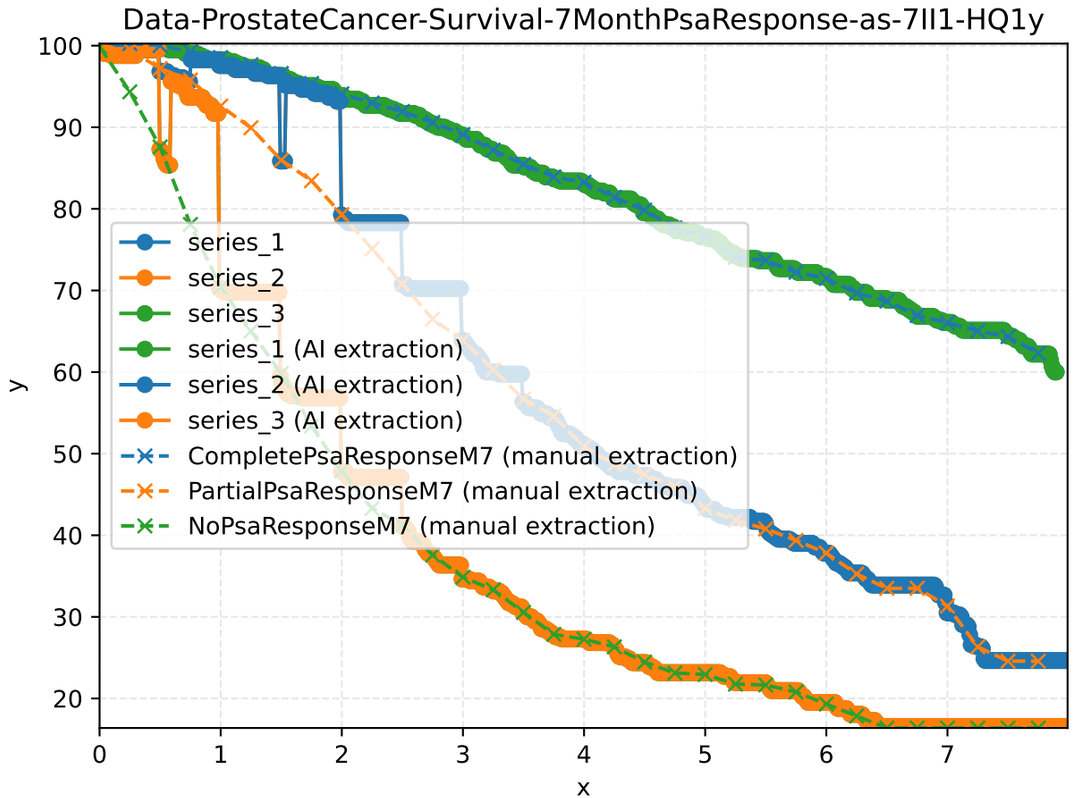


Metrics

Metric	Value
duration (s)	37
overall_nmae	0.01465
overall_integral_l1	0.693
overall_integral_l1_normalized	—
overall_x_misaligned	False
extracted_series_count	3
GT_series_count	3

Source image

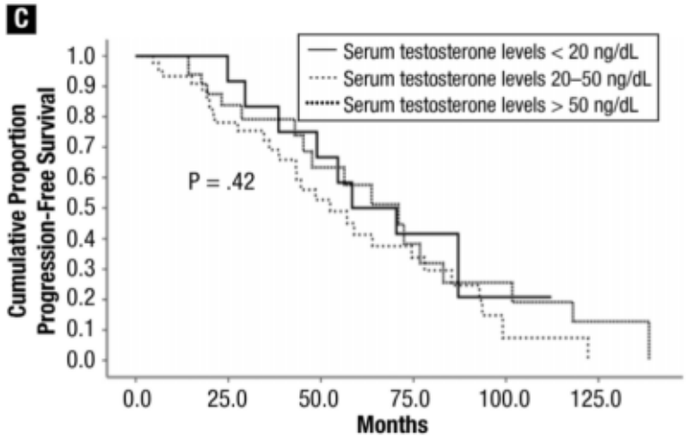




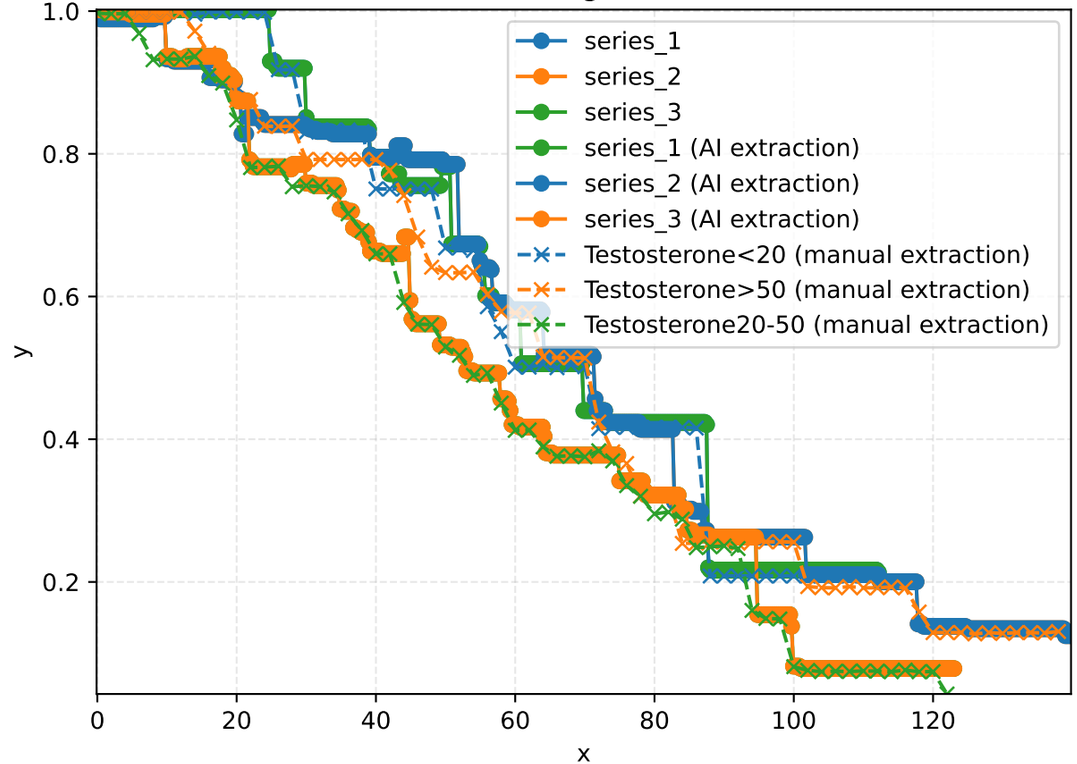
Metrics

Metric	Value
duration (s)	47
overall_nmae	0.0194
overall_integral_l1	11.01
overall_integral_l1_normalized	—
overall_x_misaligned	False
extracted_series_count	3
GT_series_count	3

Source image



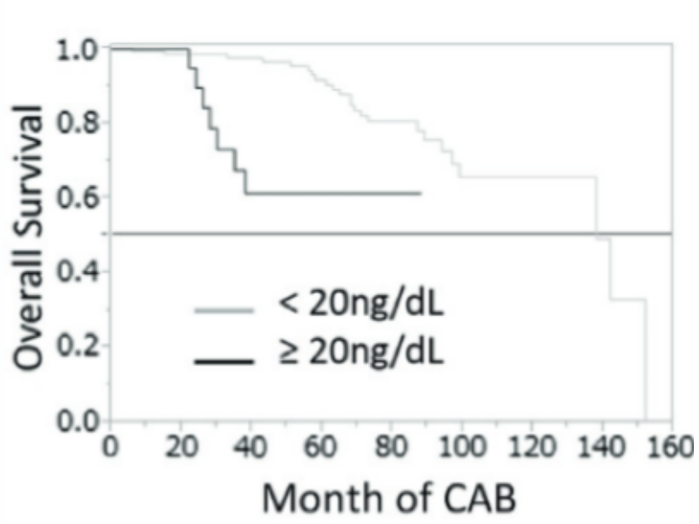
Data-ProstateCancer-CastrationLevel-TimeToProgression-BiochemicalRecurrence-as-4dag-RoWD



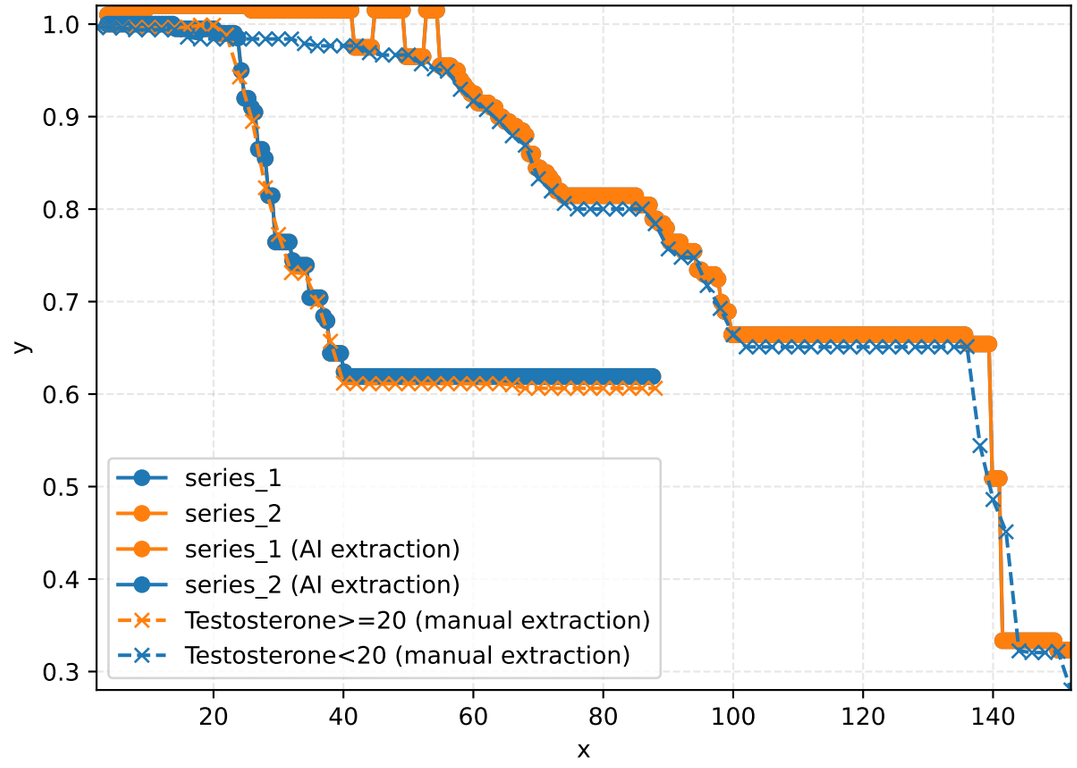
Metrics

Metric	Value
duration (s)	63
overall_nmae	0.02124
overall_integral_l1	2.348
overall_integral_l1_normalized	—
overall_x_misaligned	False
extracted_series_count	3
GT_series_count	3

Source image



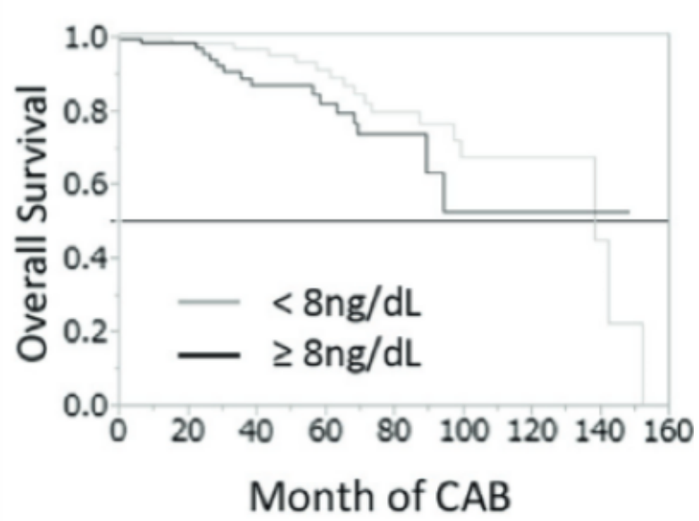
Data-ProstateCancer-OverallSurvival-SerumTestosteroneNadir-20ngPerDl-as-j2Uo-EdA3



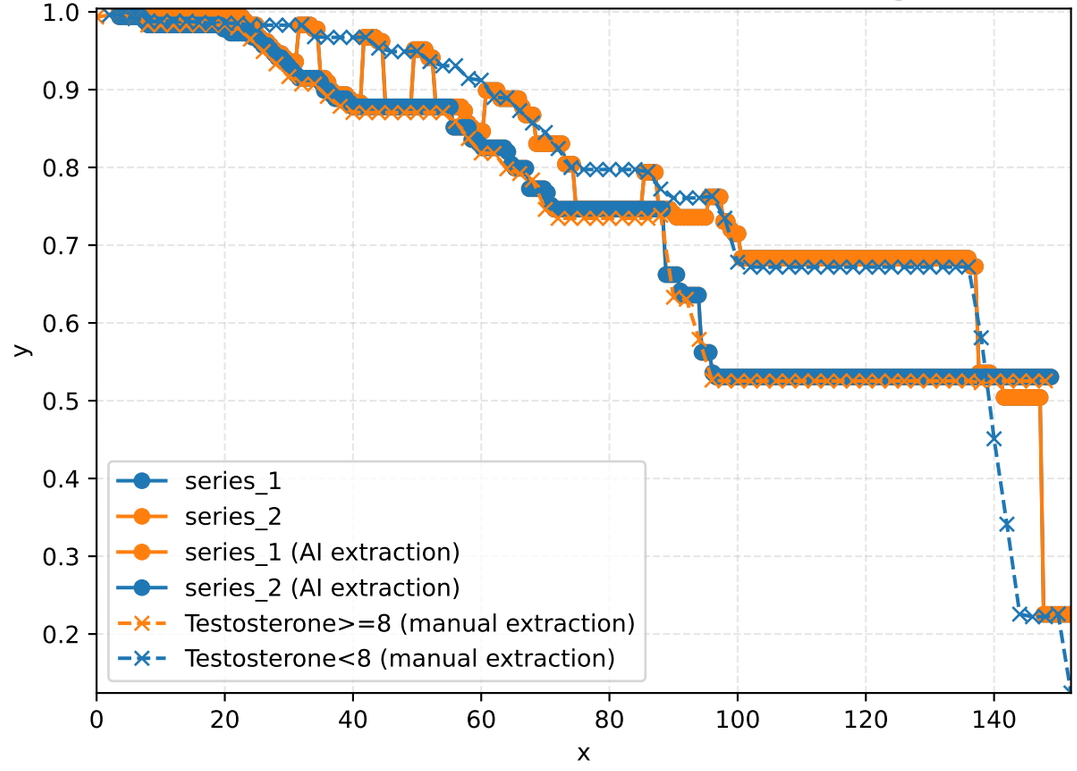
Metrics

Metric	Value
duration (s)	25
overall_nmae	0.02645
overall_integral_l1	1.978
overall_integral_l1_normalized	—
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



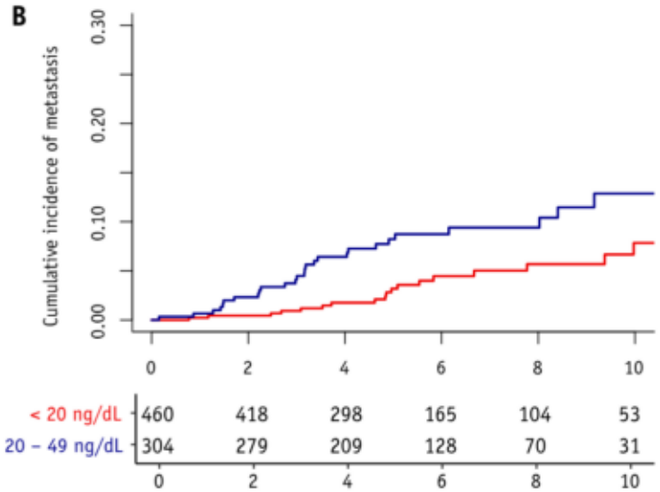
Data-ProstateCancer-OverallSurvival-SerumTestosteroneNadir-8ngPerDl-as-mu6F-pD1H

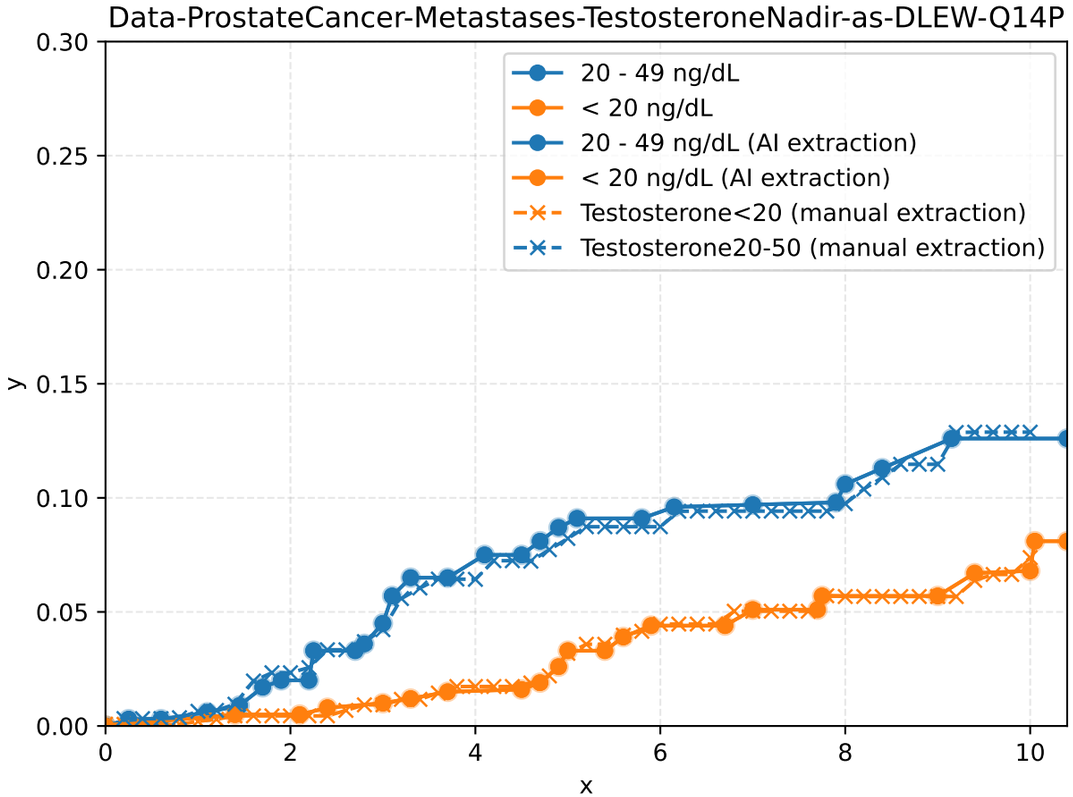


Metrics

Metric	Value
duration (s)	32
overall_nmae	0.02706
overall_integral_l1	2.976
overall_integral_l1_normalized	—
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image

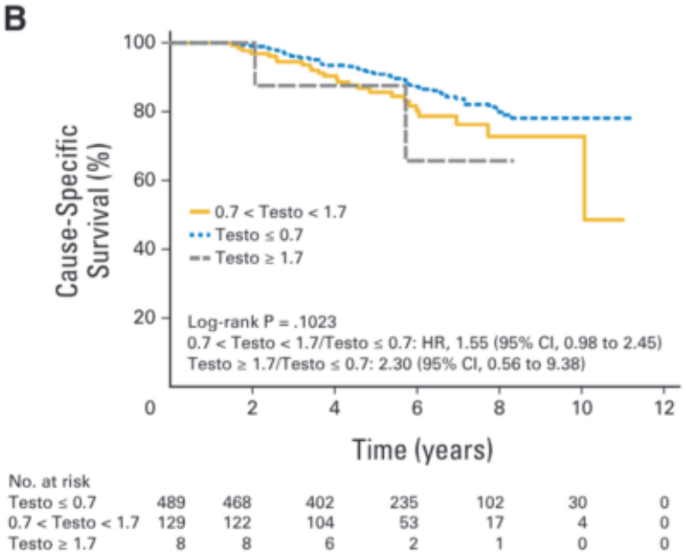




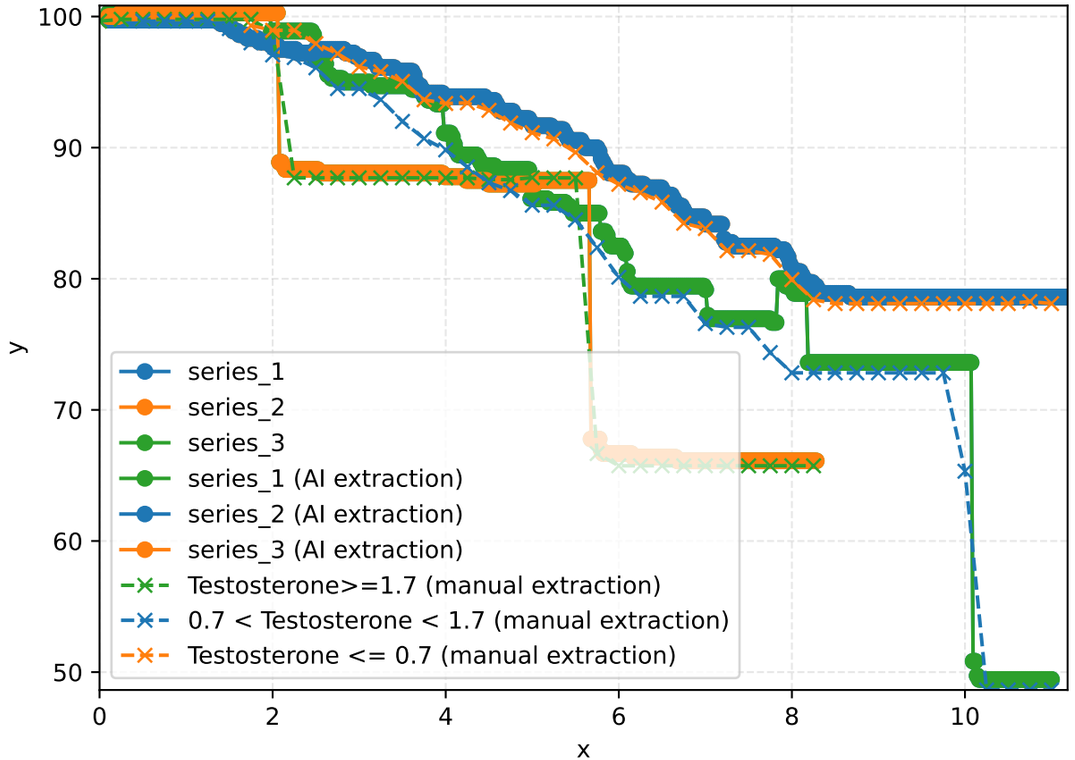
Metrics

Metric	Value
duration (s)	50
overall_nmae	0.02757
overall_integral_l1	0.02155
overall_integral_l1_normalized	0.02068
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



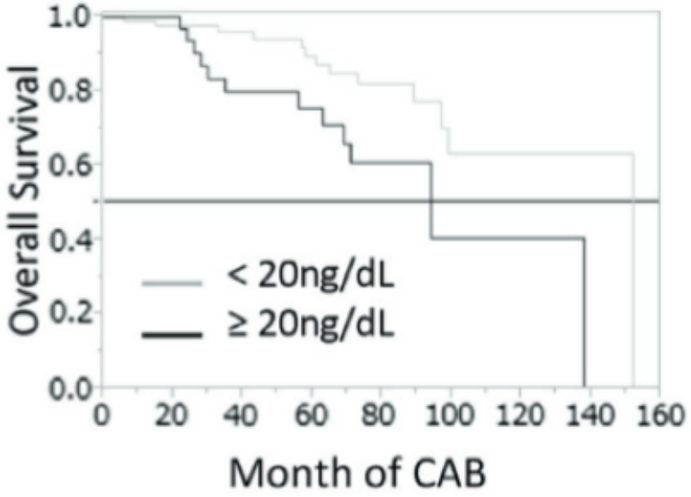
Data-ProstateCancer-TimeToCauseSpecificSurvival-TestosteroneNadir-as-zQ4P-rWVB



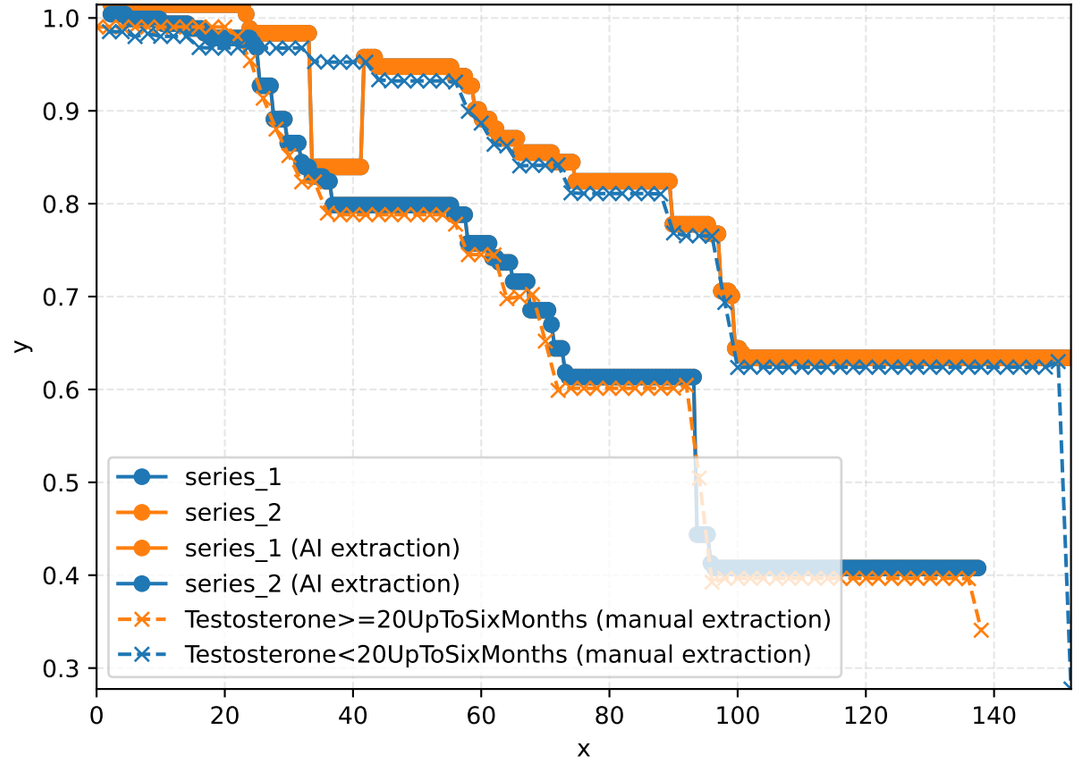
Metrics

Metric	Value
duration (s)	64
overall_nmae	0.02851
overall_integral_l1	9.879
overall_integral_l1_normalized	—
overall_x_misaligned	False
extracted_series_count	3
GT_series_count	3

Source image



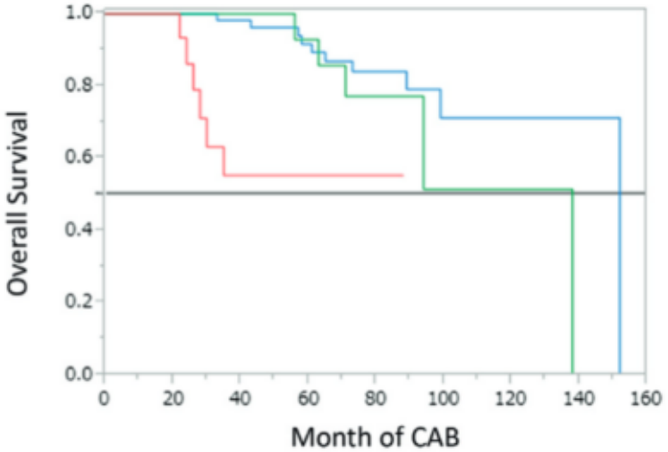
Data-ProstateCancer-OverallSurvival-SerumTestosteroneNadir-20ngPerDl-SixMonths-as-Byus-Y7YM



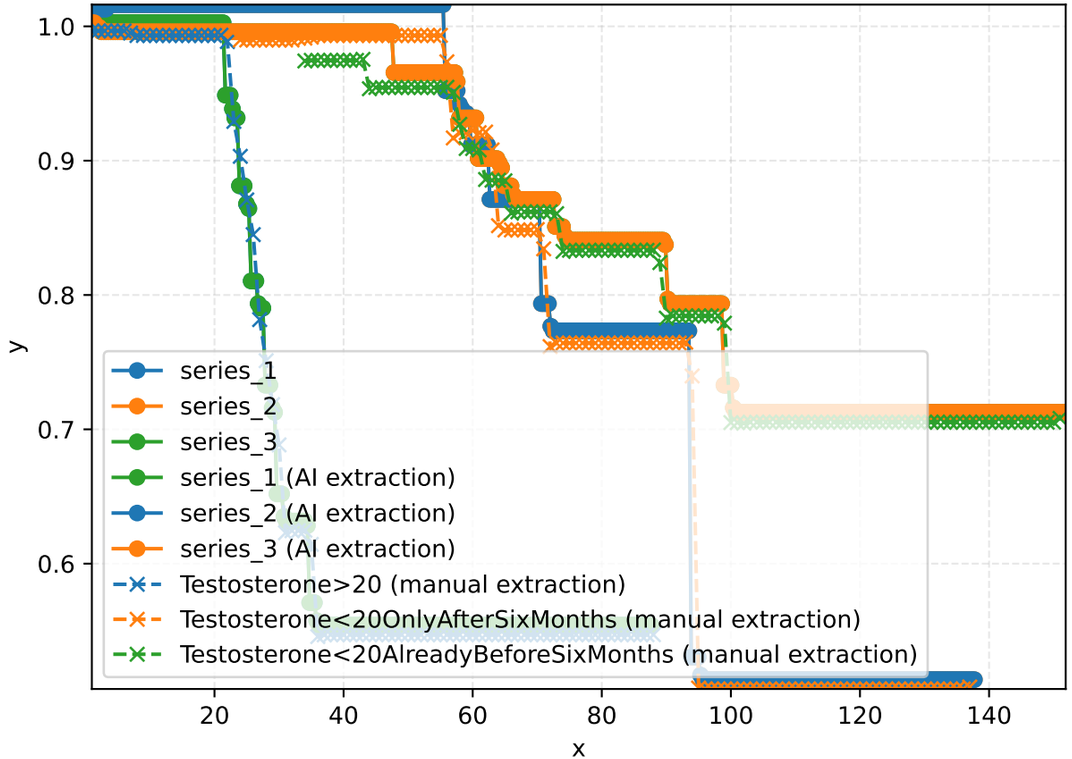
Metrics

Metric	Value
duration (s)	32
overall_nmae	0.02867
overall_integral_l1	2.675
overall_integral_l1_normalized	—
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



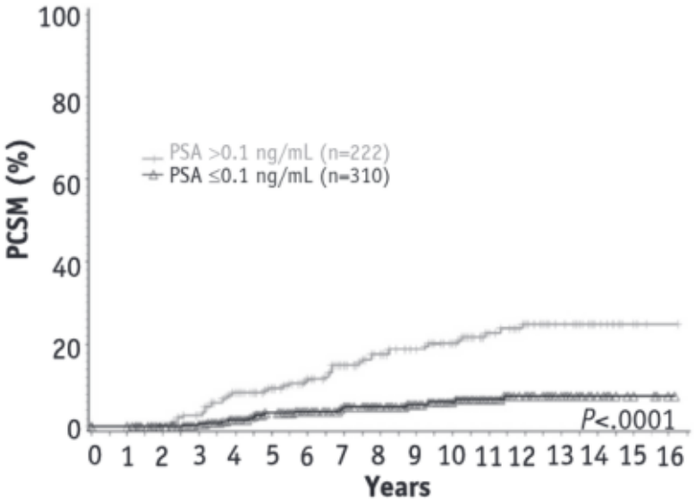
Data-ProstateCancer-OverallSurvival-TestosteroneResponseNadir-as-7x4S-GPr7



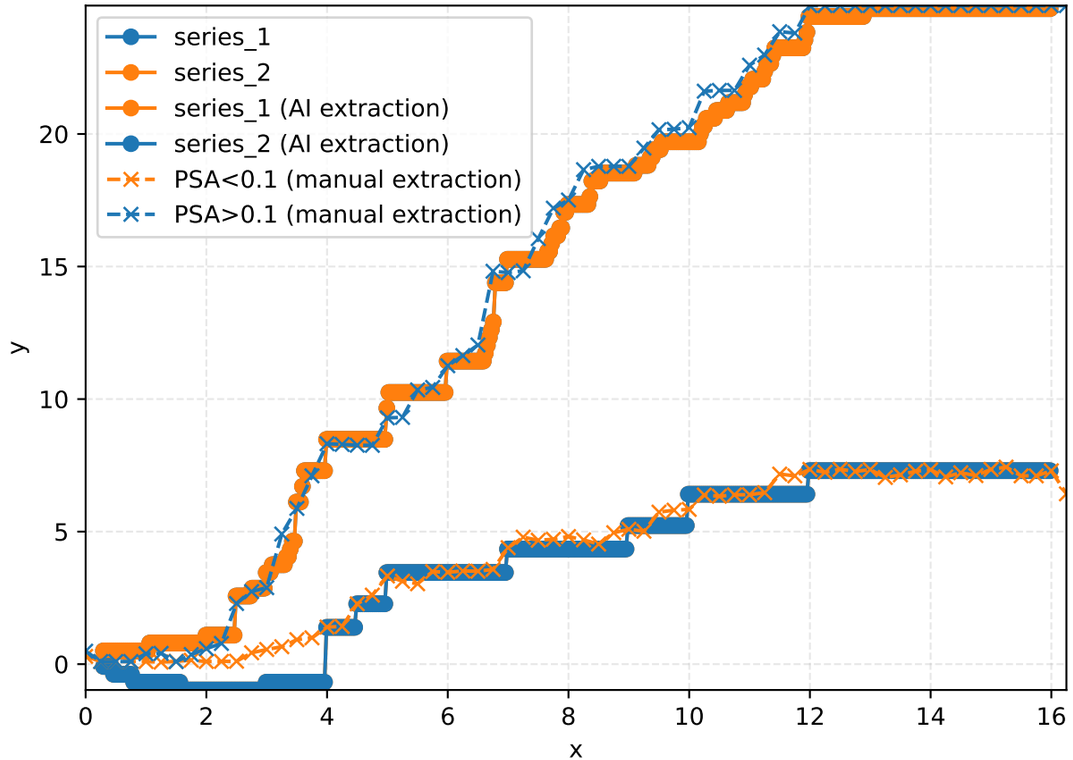
Metrics

Metric	Value
duration (s)	34
overall_nmae	0.02959
overall_integral_l1	1.279
overall_integral_l1_normalized	—
overall_x_misaligned	False
extracted_series_count	3
GT_series_count	3

Source image



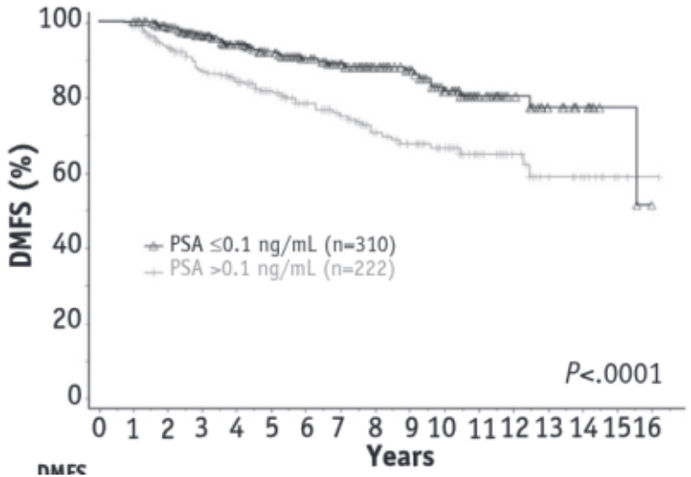
Data-Naik2016-6MonthPSA-Strat-ProstateCancerSpecificMortality-as-AXhW-IH9H



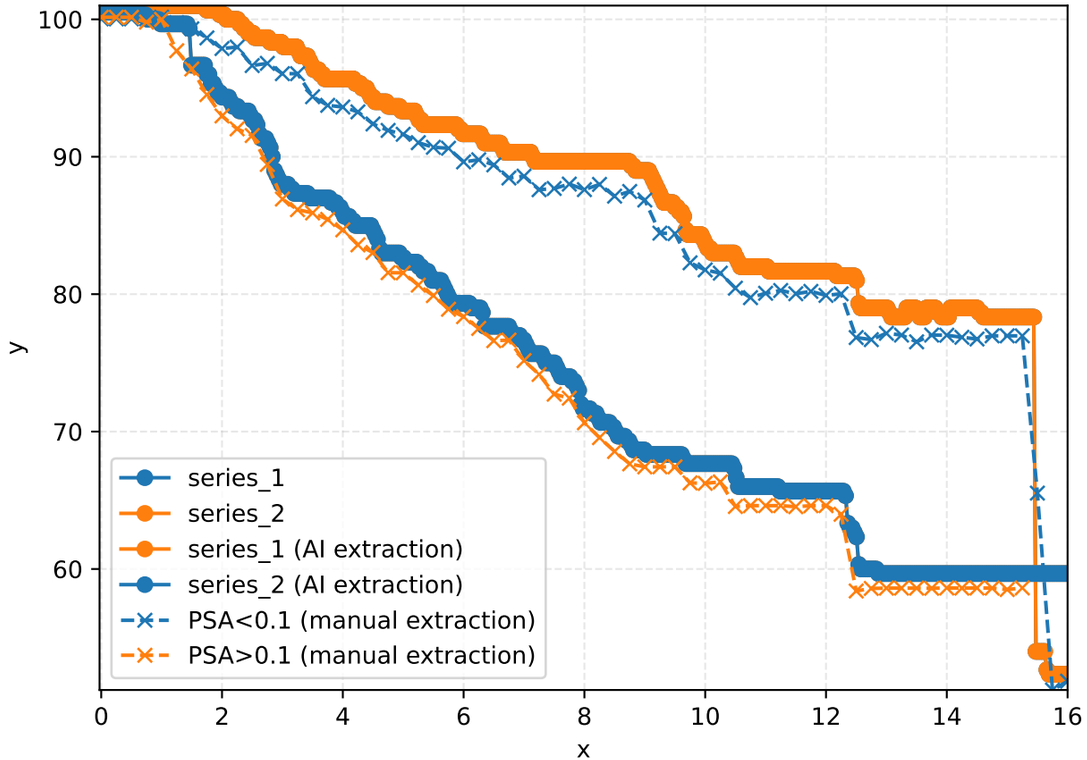
Metrics

Metric	Value
duration (s)	31
overall_nmae	0.03437
overall_integral_l1	6.819
overall_integral_l1_normalized	—
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



Data-Naik2016-6MonthPSA-Strat-MetastasisFreeSurvival-as-cLMA-G8dv

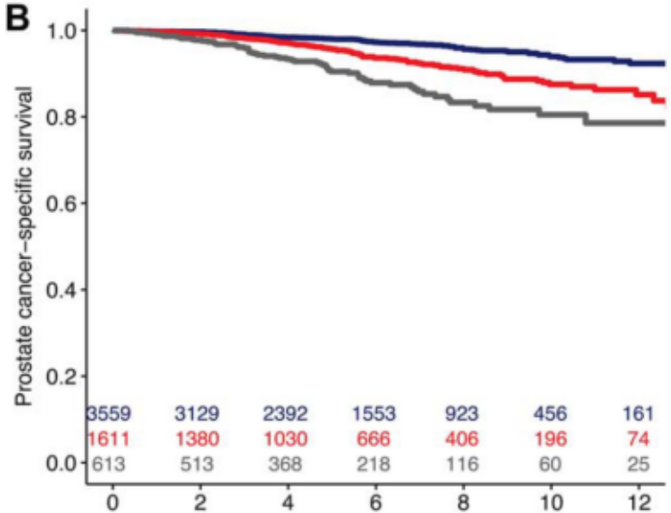


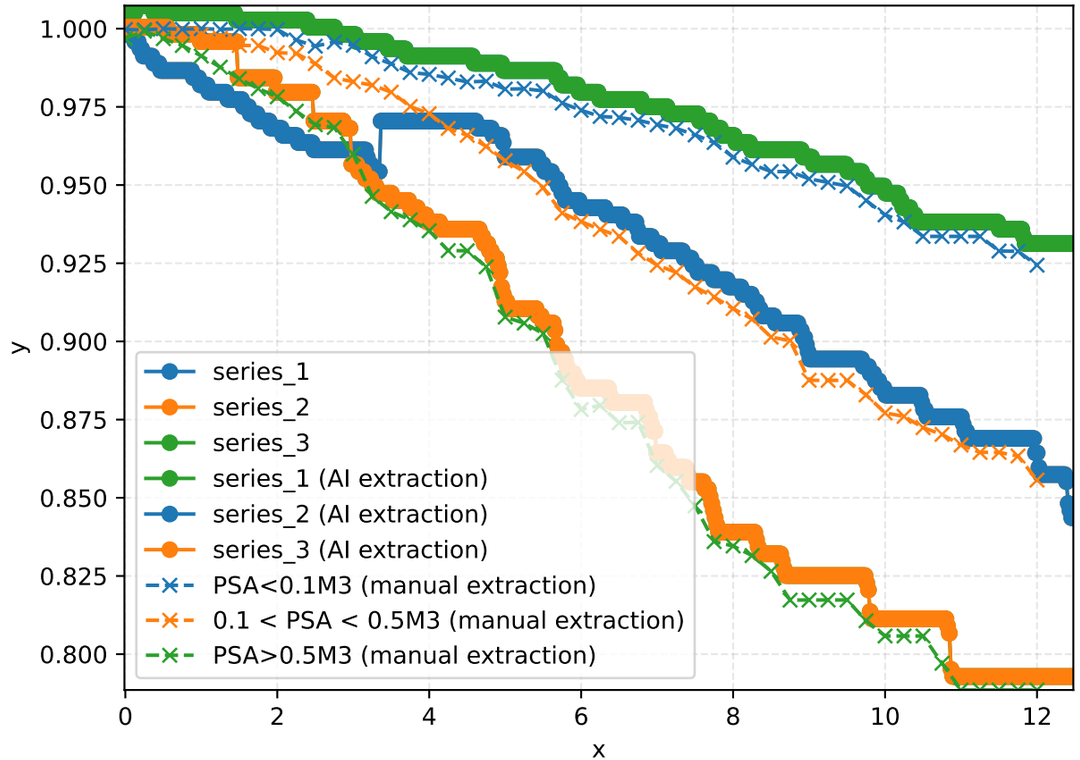
Metrics

Metric	Value
duration (s)	31
overall_nmae	0.03455
overall_integral_l1	25.24
overall_integral_l1_normalized	—
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Data-ProstateCancer-DiseaseSpecificMortality-FirstLineLocalized-Radiotherapy-Bryant2018-as-PBP9-LpbG

Source image

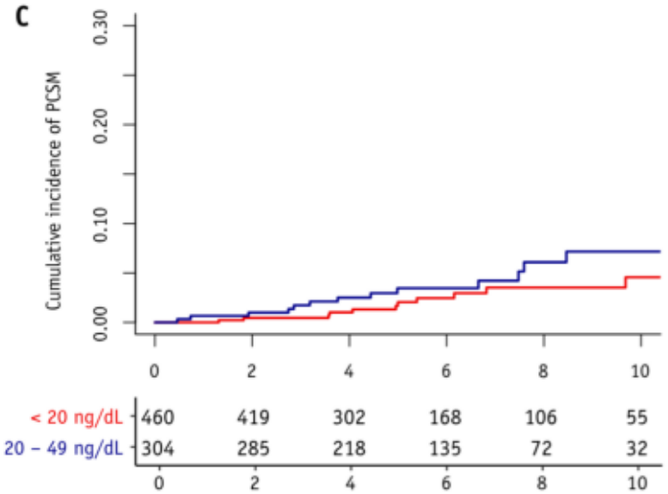




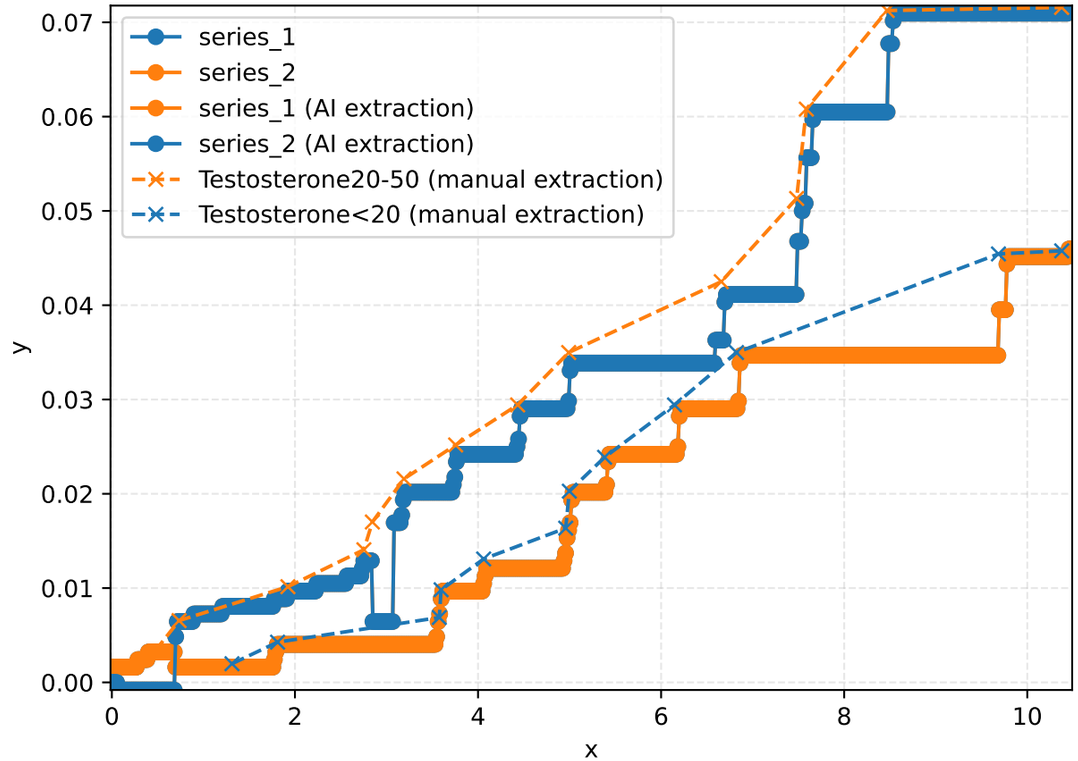
Metrics

Metric	Value
duration (s)	88
overall_nmae	0.05503
overall_integral_l1	0.08143
overall_integral_l1_normalized	—
overall_x_misaligned	False
extracted_series_count	3
GT_series_count	3

Source image



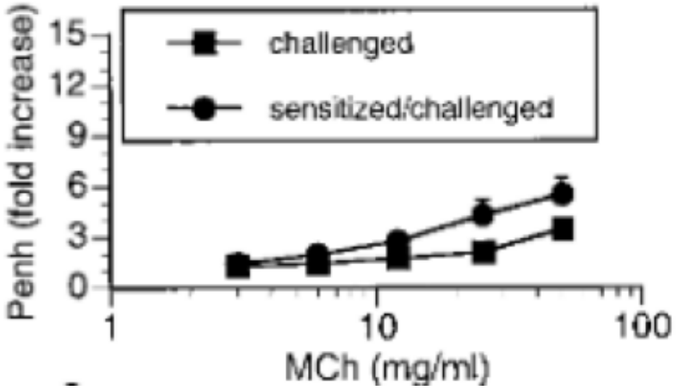
Data-ProstateCancer-ProstateCancerSpecificMortality-TestosteroneNadir-as-JAS0-KOng



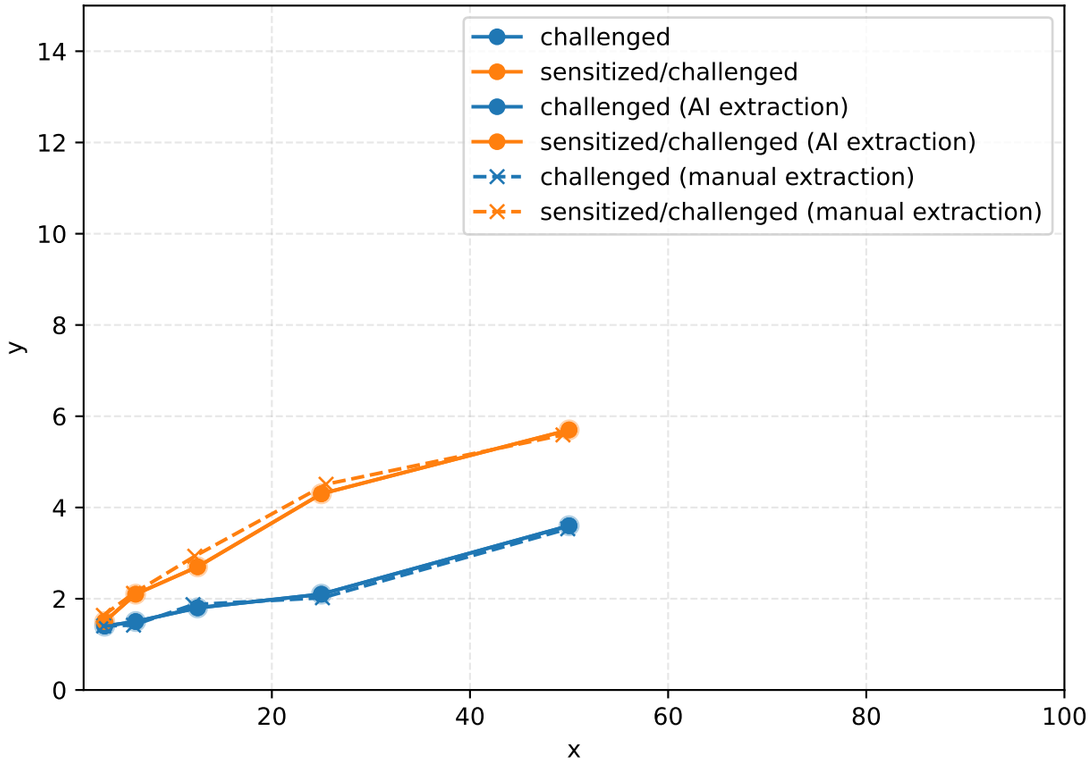
Metrics

Metric	Value
duration (s)	41
overall_nmae	0.05842
overall_integral_l1	0.03054
overall_integral_l1_normalized	—
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



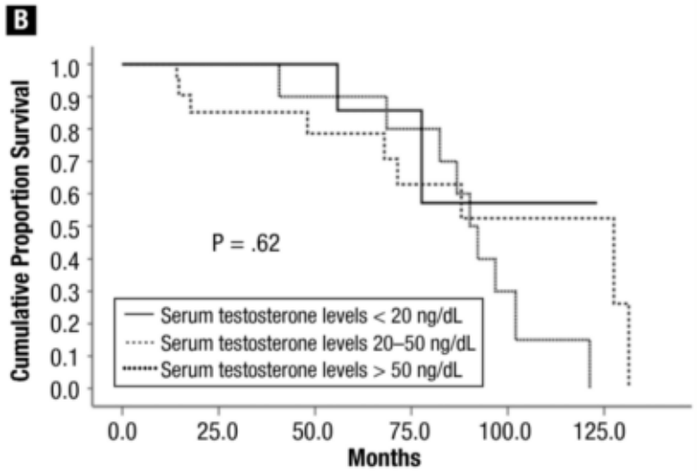
Data-as-CdWI-s6OK



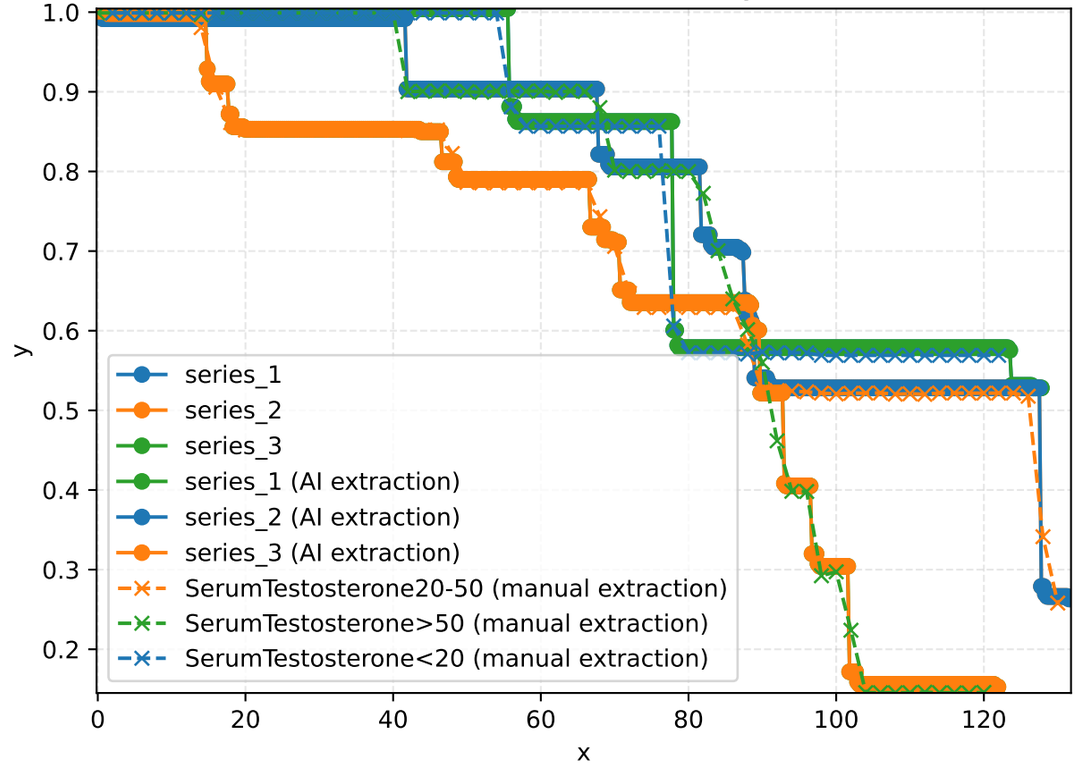
Metrics

Metric	Value
duration (s)	22
overall_nmae	0.06607
overall_integral_l1	2.713
overall_integral_l1_normalized	0.0183
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



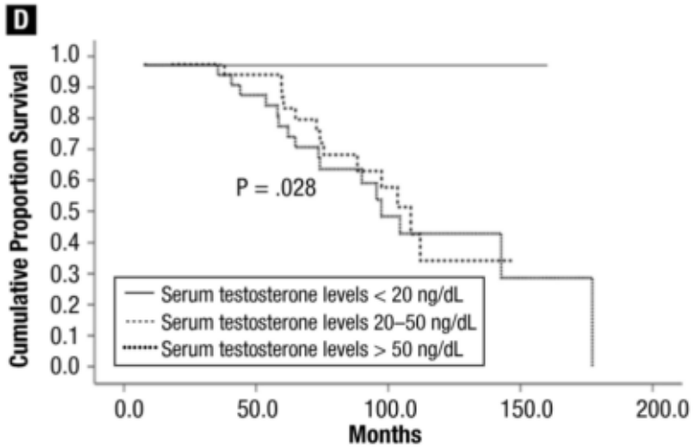
Data-ProstateCancer-CastrationLevel-OverallSurvival-LocallyAdvanced-Metastatic-as-HN9j-M4NK



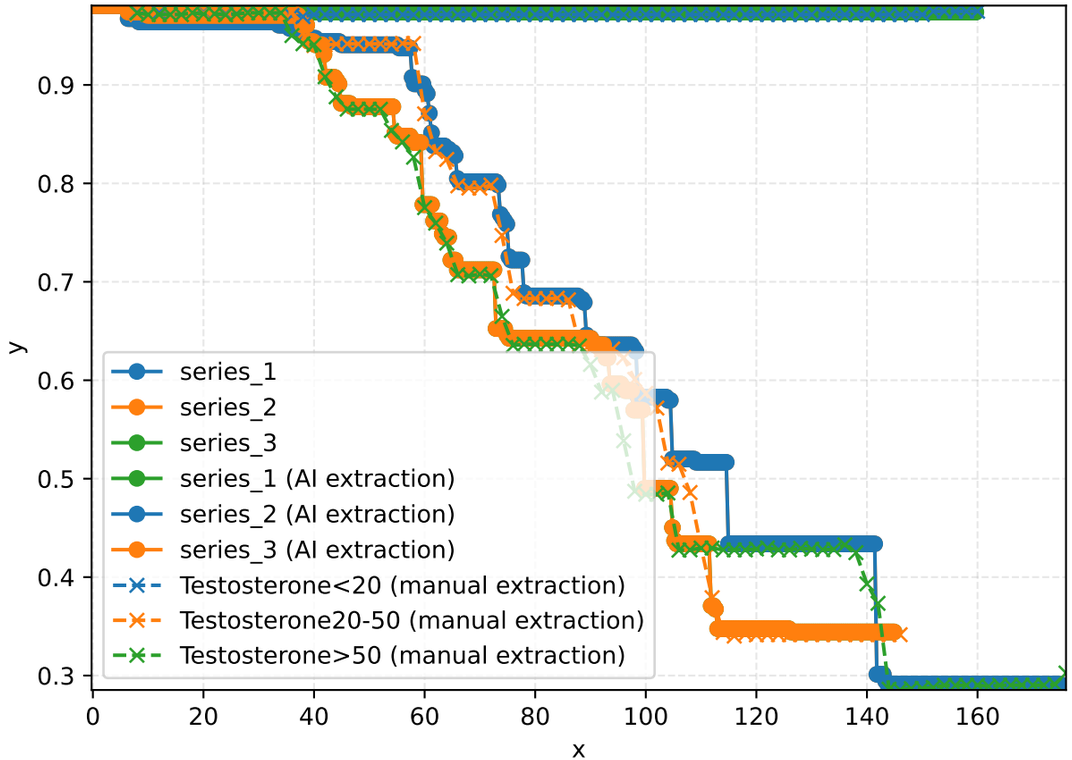
Metrics

Metric	Value
duration (s)	54
overall_nmae	0.07023
overall_integral_l1	6.641
overall_integral_l1_normalized	—
overall_x_misaligned	False
extracted_series_count	3
GT_series_count	3

Source image



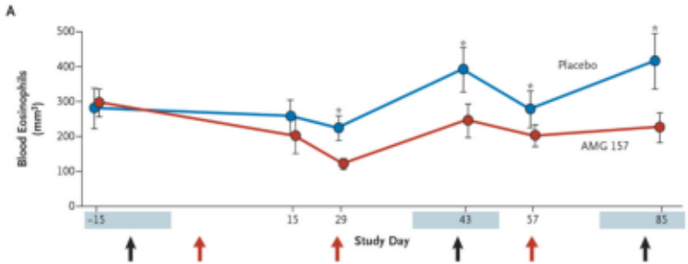
Data-ProstateCancer-CastrationLevel-OverallSurvival-BiochemicalRecurrence-as-KBNL-4YCX



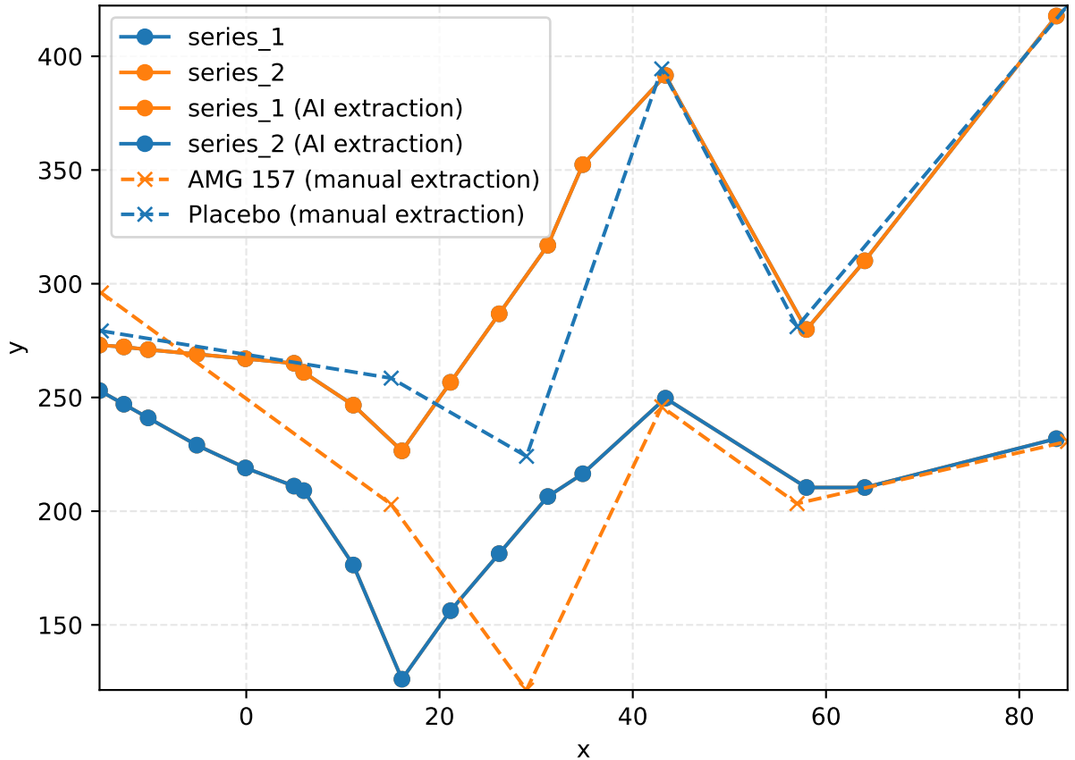
Metrics

Metric	Value
duration (s)	44
overall_nmae	0.1168
overall_integral_l1	2.826
overall_integral_l1_normalized	—
overall_x_misaligned	False
extracted_series_count	3
GT_series_count	3

Source image



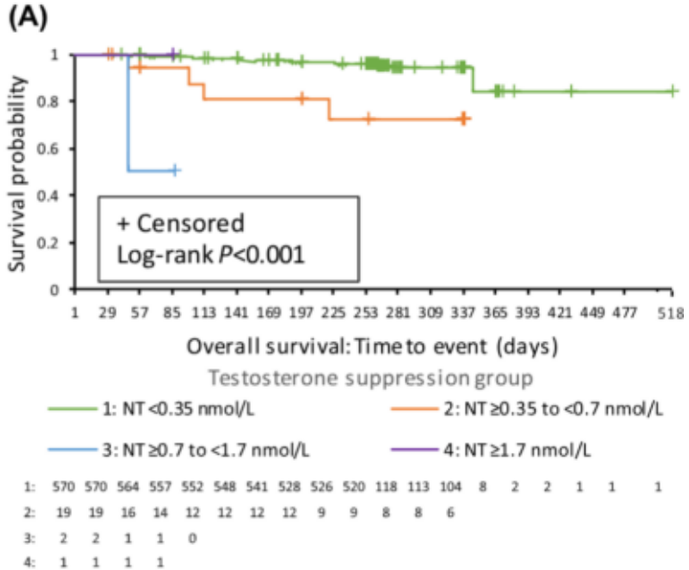
Data-as-8IYI-AGTa



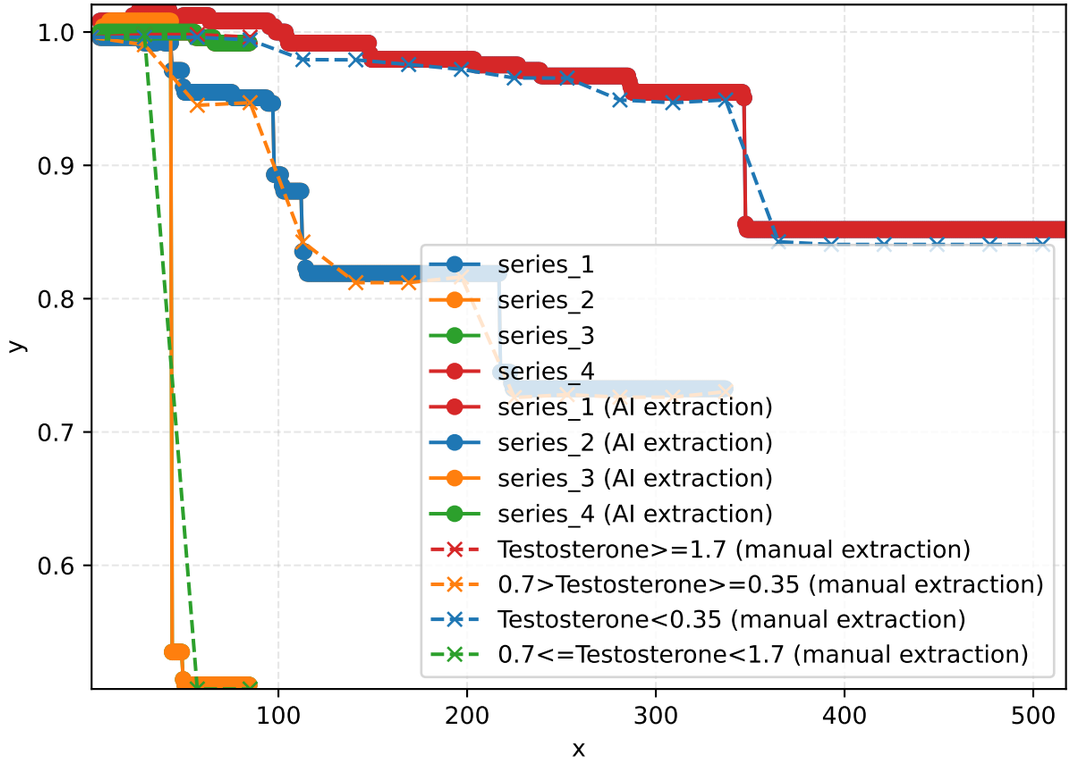
Metrics

Metric	Value
duration (s)	35
overall_nmae	0.1363
overall_integral_l1	1812
overall_integral_l1_normalized	—
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



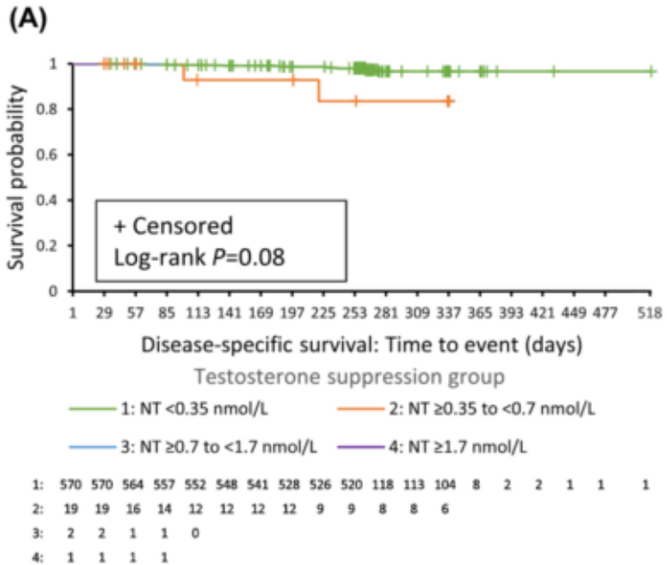
Data-ProstateCancer-OverallSurvival-TestosteroneSuppression-as-gIUH-EPAn



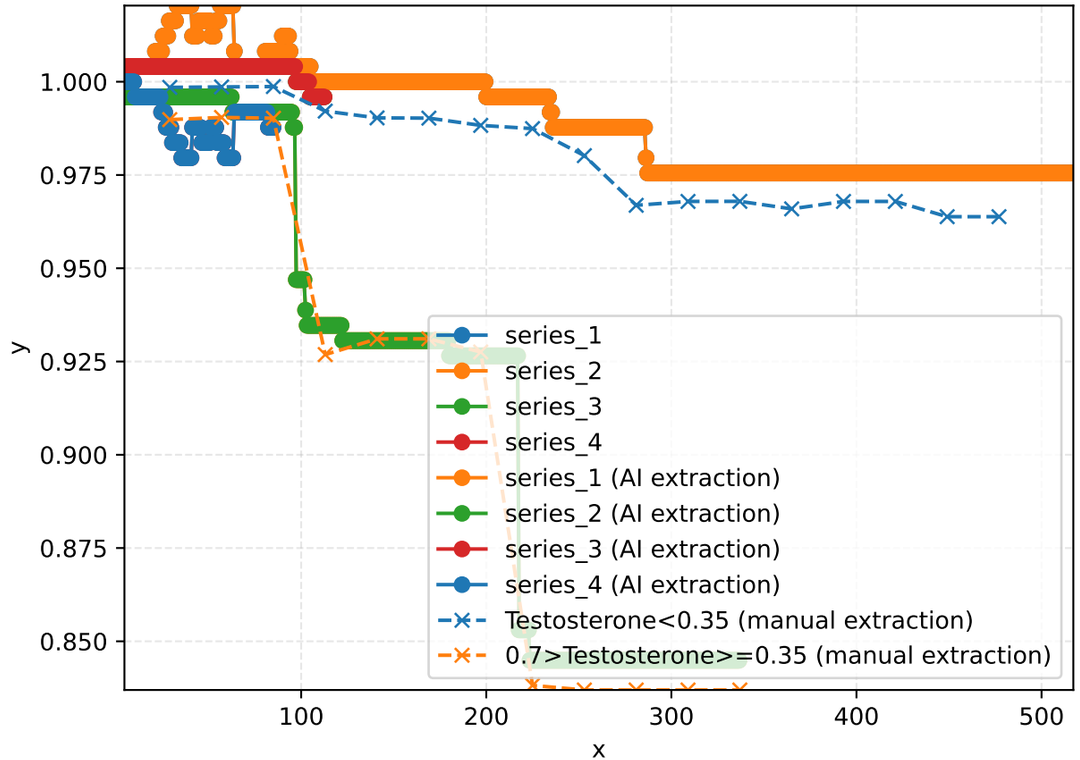
Metrics

Metric	Value
duration (s)	123
overall_nmae	0.1435
overall_integral_l1	3.071
overall_integral_l1_normalized	—
overall_x_misaligned	False
extracted_series_count	4
GT_series_count	4

Source image



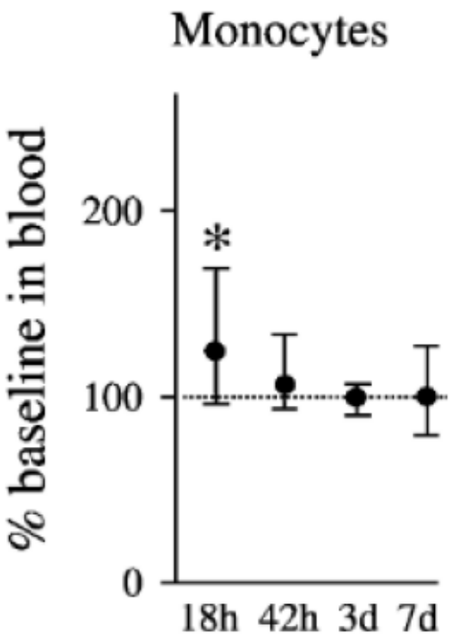
Data-ProstateCancer-DiseaseSpecificSurvival-TestosteroneSuppression-as-pk2y-g1My



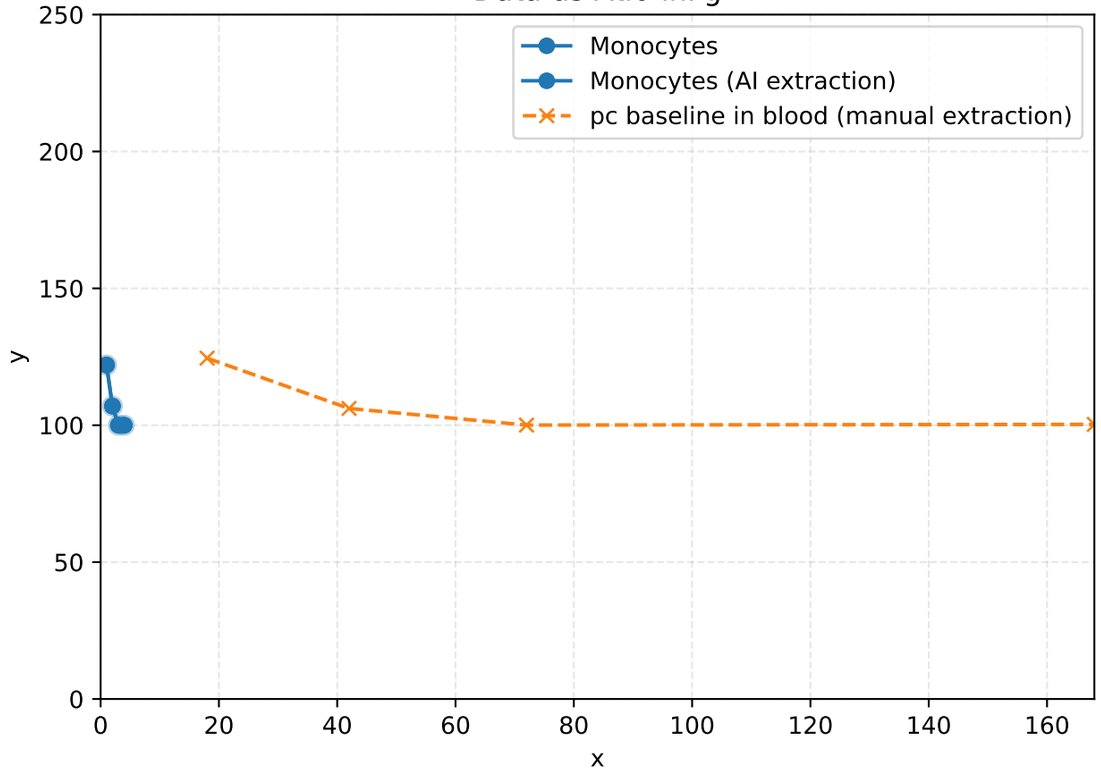
Metrics

Metric	Value
duration (s)	268
overall_nmae	0.4205
overall_integral_l1	0.3539
overall_integral_l1_normalized	—
overall_x_misaligned	True
extracted_series_count	4
GT_series_count	2

Source image



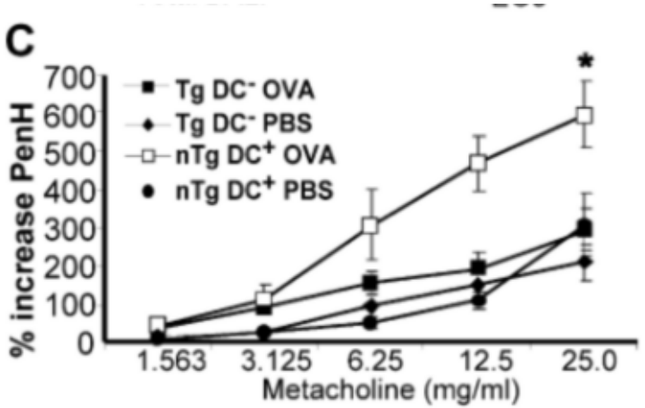
Data-as-Atl6-ihFg



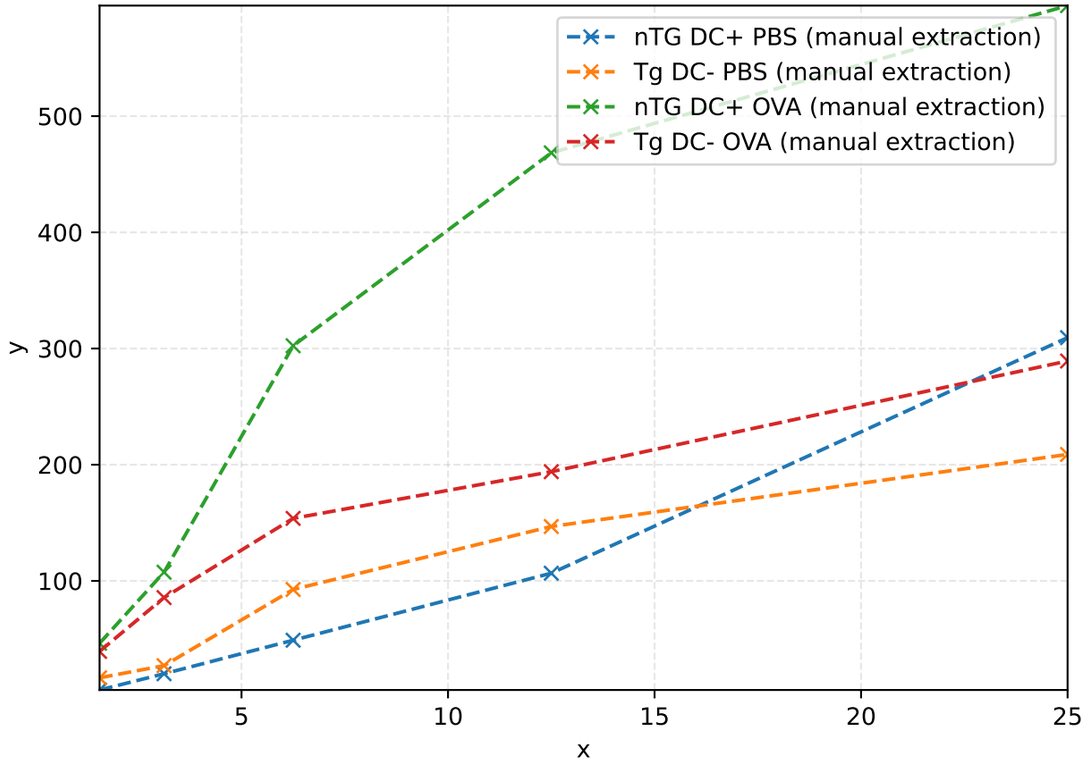
Metrics

Metric	Value
duration (s)	43
overall_nmae	nan
overall_integral_l1	nan
overall_integral_l1_normalized	—
overall_x_misaligned	True
extracted_series_count	1
GT_series_count	1

Source image



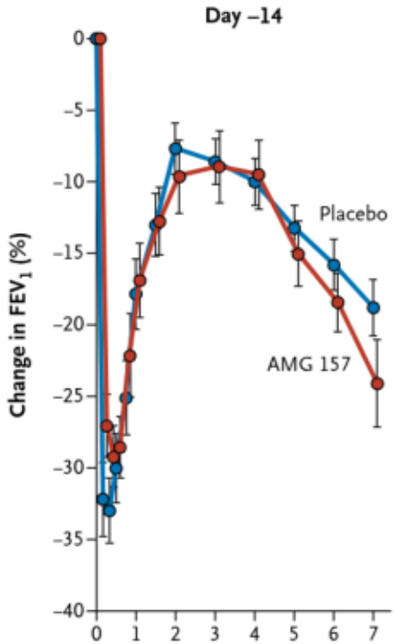
Data-as-YpSF-UNAQ



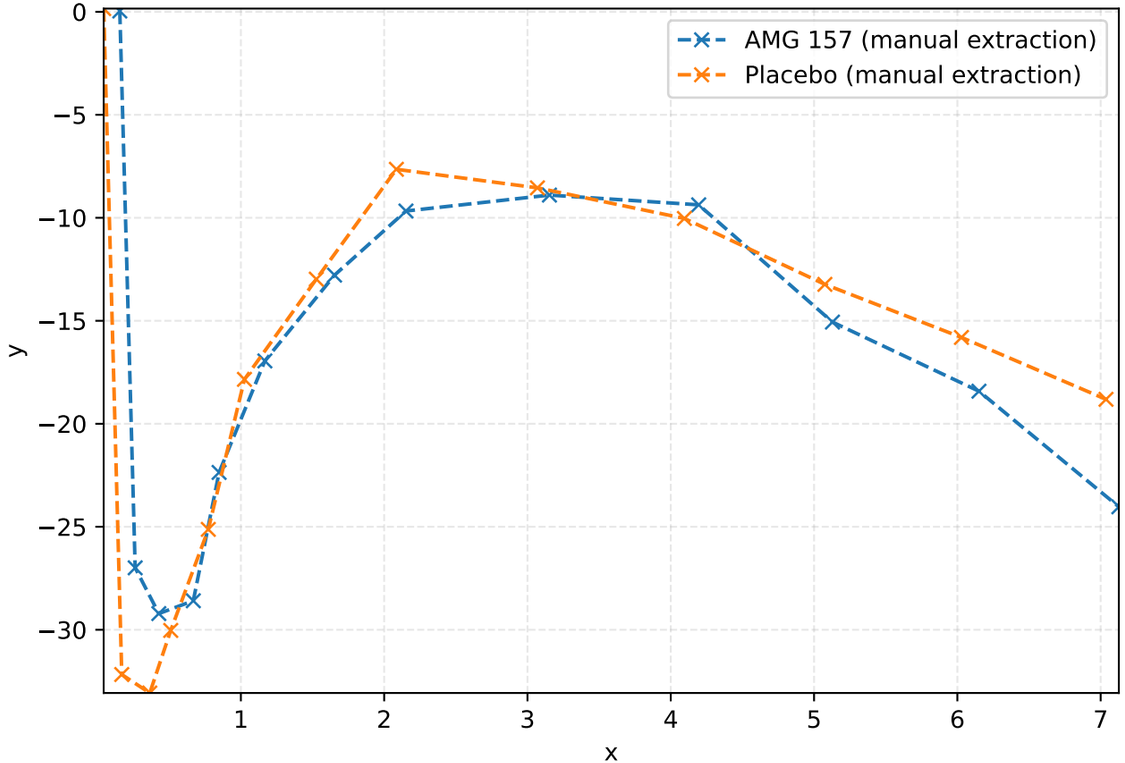
Metrics

Metric	Value
duration (s)	—
overall_nmae	—
overall_integral_l1	—
overall_integral_l1_normalized	—
overall_x_misaligned	—
extracted_series_count	—
GT_series_count	—

Source image



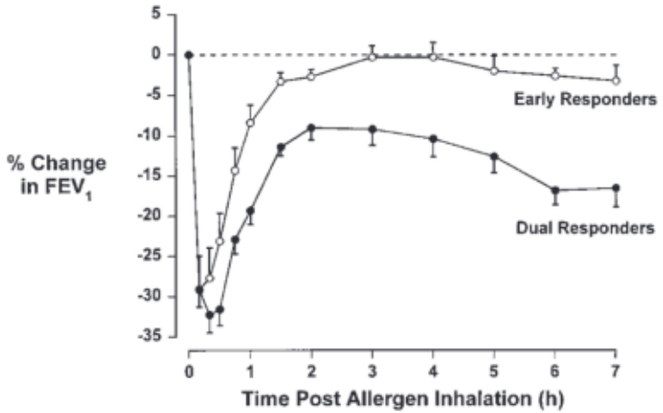
Data-as-b98m-TsDT



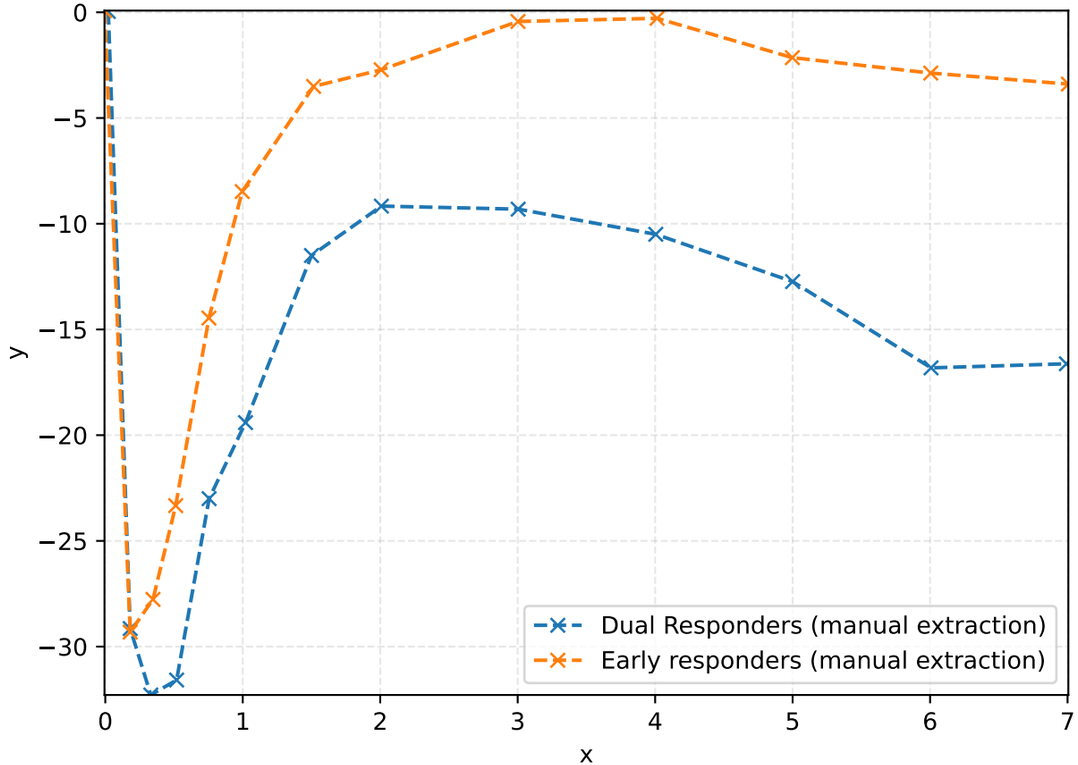
Metrics

Metric	Value
duration (s)	—
overall_nmae	—
overall_integral_l1	—
overall_integral_l1_normalized	—
overall_x_misaligned	—
extracted_series_count	—
GT_series_count	—

Source image



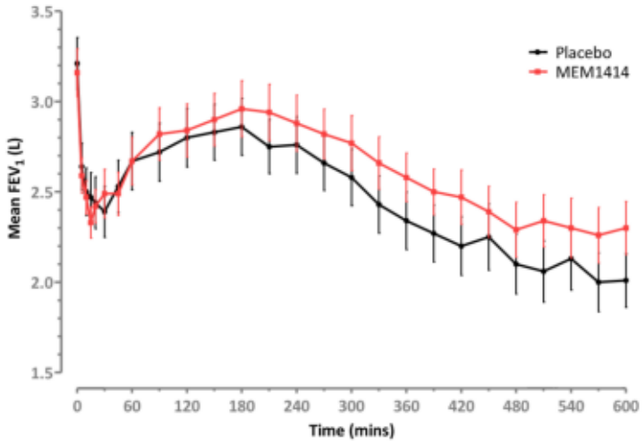
Data-as-bU4o-xO6R



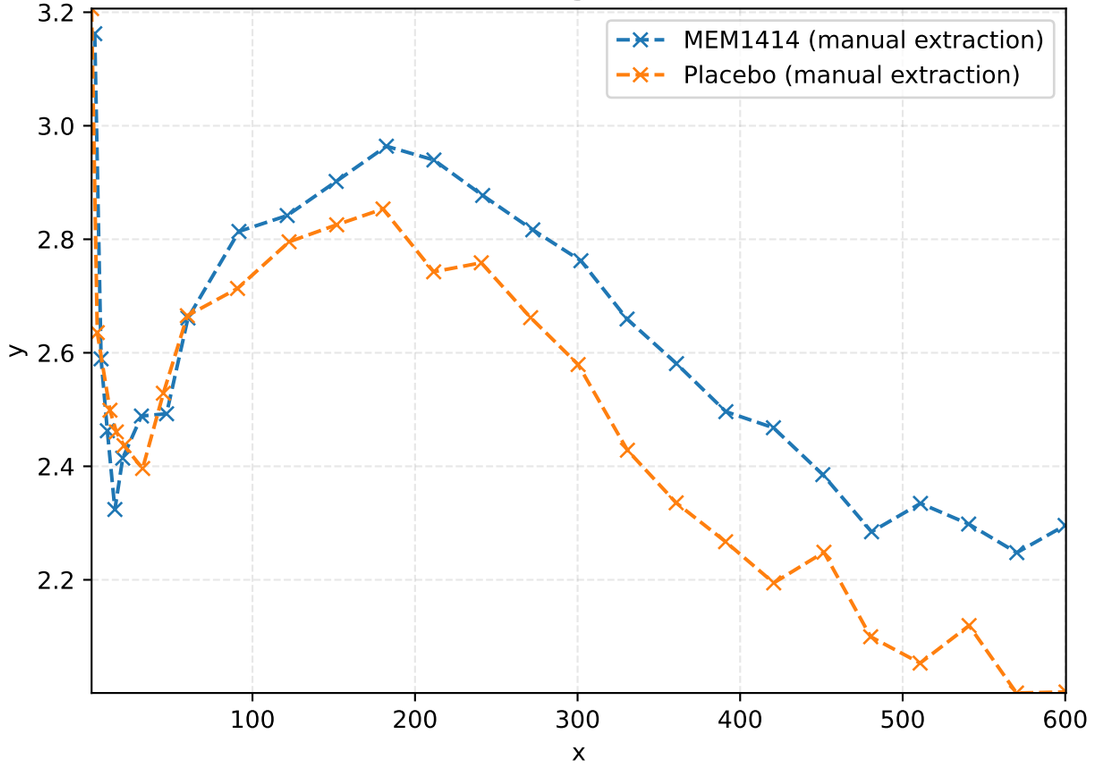
Metrics

Metric	Value
duration (s)	—
overall_nmae	—
overall_integral_l1	—
overall_integral_l1_normalized	—
overall_x_misaligned	—
extracted_series_count	—
GT_series_count	—

Source image



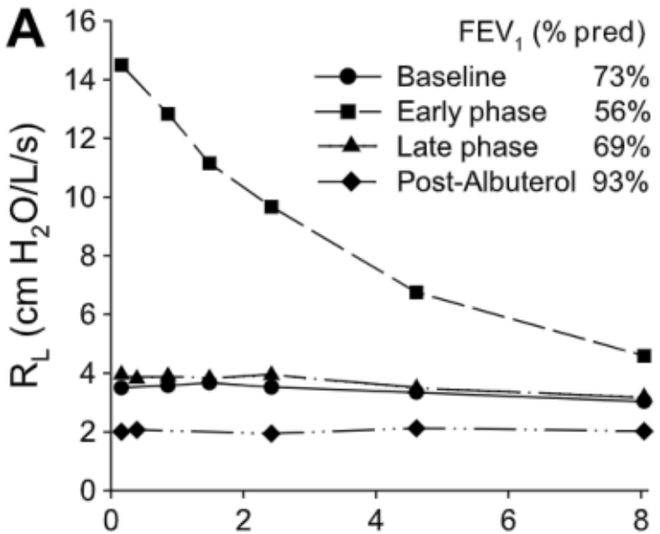
Data-as-ngkV-QcWm



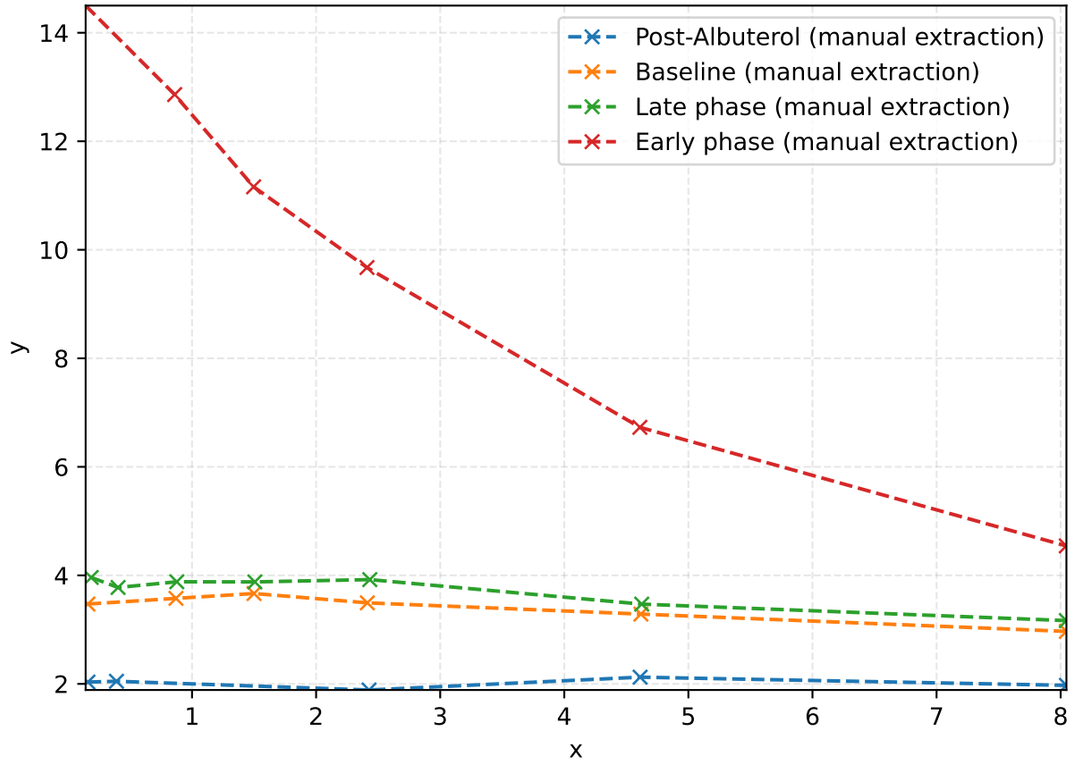
Metrics

Metric	Value
duration (s)	—
overall_nmae	—
overall_integral_l1	—
overall_integral_l1_normalized	—
overall_x_misaligned	—
extracted_series_count	—
GT_series_count	—

Source image



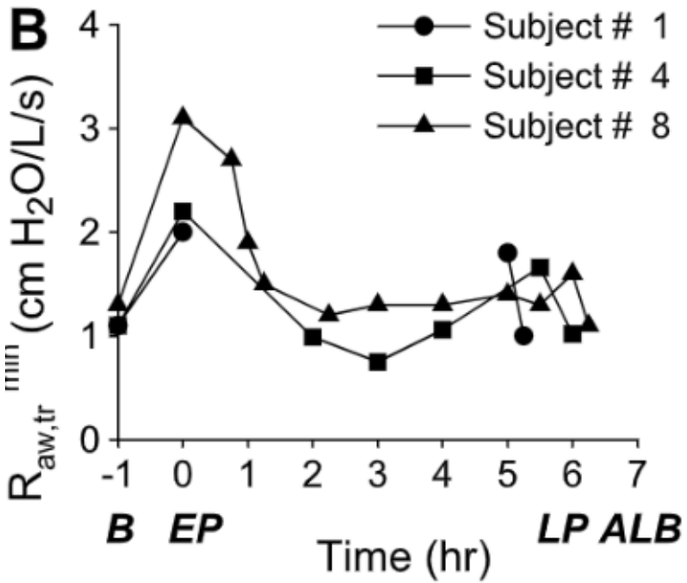
Data-as-y2nv-TPY1



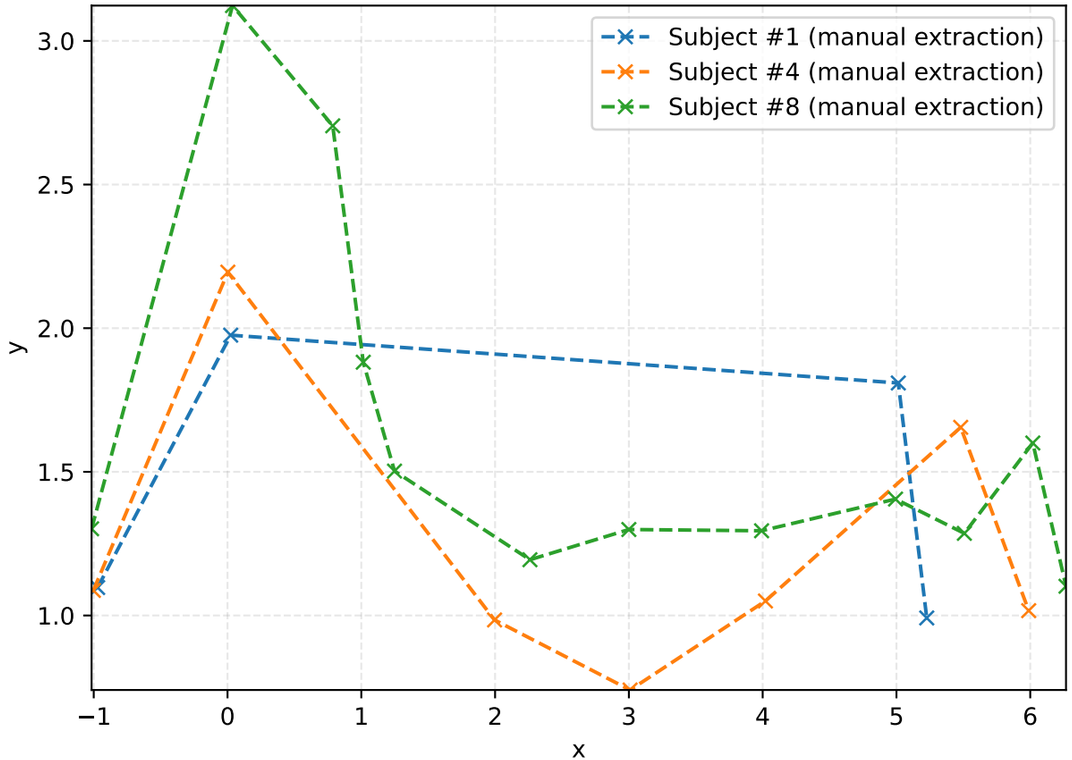
Metrics

Metric	Value
duration (s)	—
overall_nmae	—
overall_integral_l1	—
overall_integral_l1_normalized	—
overall_x_misaligned	—
extracted_series_count	—
GT_series_count	—

Source image



Data-as-ylJc-Vi0h



Metrics

Metric	Value
duration (s)	—
overall_nmae	—
overall_integral_l1	—
overall_integral_l1_normalized	—
overall_x_misaligned	—
extracted_series_count	—
GT_series_count	—